



A review article based on composite graphene @tungsten oxide thin films for various applications

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Received: 6 April 2022 / Revised: 20 May 2022 / Accepted: 26 May 2022 / Published online: 12 August 2022
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Abstract

Graphene and its derivatives are the hot topics of research during this decade due to their excellent thermal conductivities, mechanical strength, current densities, electron motilities, and large surface area. This review article explores the outstanding applicability and features of graphene derivatives. The transition metal oxides (TMOs) have also gained considerable research attention due to their unique physicochemical properties in photocatalytic, self-cleaning, and gas sensing applications. Among TMOs, tungsten metal oxides have received a tremendous response as they are naturally abundant, low in cost, less toxic, environmental friendly, and can be manufactured using various physical and chemical methods. It exhibits a cubic perovskite-like structure based on the corner-sharing of regular octahedra with the oxygen atoms at the corner and the tungsten atoms at the centre of each octahedron. It also shows structural polymorphism and substoichiometric phase transitions, which attracted the attention of researchers over the past few years to explore their potential in various applications. Pairing graphene and its derivatives with tungsten oxide (WO_3) to create heterojunction could be an auspicious tool to improve photocatalysis, energy storage, medical, electrochromism, and energy efficiency conversion. In addition, composite exhibits significantly higher efficiency than either individual material due to their well-matched band edge positions, efficient charge separation, and light-harvesting abilities. The morphology and heterojunction were found to be quite beneficial in improving the overall performance of the composite. In this review article, the noteworthy endeavors and turning points are accomplished utilizing heterojunction between WO_3 and graphene derivatives for different applications. This review article will also provide the research gap and excite new ideas for further improvement of graphene-based tungsten oxide nanocomposites. Conclusively, the scope of future research work to design the ternary composite with high efficiency utilizing WO_3 and graphene is also explored.

Keywords Tungsten oxide · Reduced graphene oxide · Composite · Photocatalytic · Gas sensor · Electrochromism

1 Introduction

Graphene, a single sheet of sp^2 hybridized carbon atoms structured in a honeycombed two-dimensional (2D) network with six-membered rings (shown in Fig. 1), has sparked interest in the field of material science [1]. Stimulating properties of graphene are attributed to its unique plane structure and geometry, which include high Young's

modulus (1100 GPa), high fracture strength (125 GPa), excellent electrical ($106 \text{ S}\cdot\text{cm}^{-1}$) and thermal conductivity ($5000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$), fast charge carrier mobility ($200,000 \text{ cm}^2\cdot(\text{V}\cdot\text{s})^{-1}$), and large specific surface area (theoretically calculated value, $2630 \text{ m}^2\cdot\text{g}^{-1}$) [2, 3]. These remarkable properties of graphene provide infinite possibilities for various applications in many areas such as electronics, energy storage and conversion, biotechnology, and especially improvement in composite fiber materials [4]. Graphene has been pronounced differently by many researchers as GR, Gr, GE, and GN, but they have the same meaning. Graphene oxide (GO, an oxidized single or multi-layered graphene) and reduced graphene oxide (rGO) is practically the most studied graphene derivatives for various applications [1], where oxygen is introduced to graphene through chemical oxidation. Moreover, GO is described as heavily

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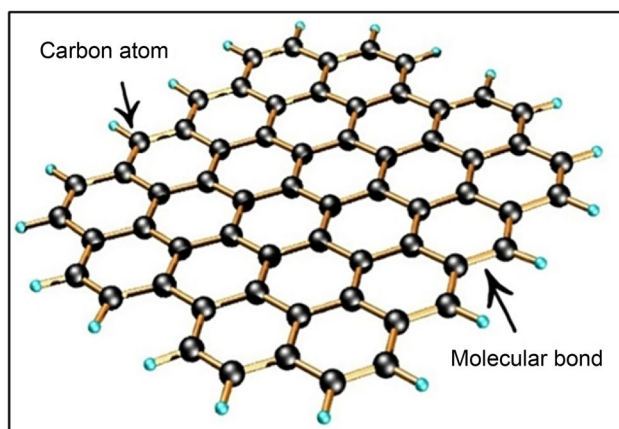


Fig. 1 Structure of one-atom-thick planar sheet of graphene

oxygenated, with the presence of many oxygen-containing functional groups such as epoxide, hydroxyl, carbonyl, and carboxyl groups on its basal plane [1]. As a result of the presence of these functional groups, GO becomes hydrophilic, allowing more significant interfacial interaction with polar polymer matrices, resulting in improved mechanical and electrical properties for a variety of applications [1]. However, the oxygen functionalization on GO reduced the electrical conductivity, and therefore, GO was less preferable for conductive polymer-based composites [5]. Besides, rGO is an intriguing graphene derivative with unique features, such as a large surface area, extraordinary charge-transfer capabilities, high conductivity, high intrinsic electron mobility, and high thermal and chemical stability [6]. Because of these properties, rGO sheets offer exciting possibilities for preparing various composites with metal-oxides, which help to enhance the efficiency in photocatalysis, electrochemical supercapacitors, electrochromic, and gas sensor applications [7–12].

On the other hand, among several transition metal oxides, tungsten oxide (WO_3), an oxygen-deficient n-type wide bandgap semiconductor material, has received much attention [13]. It is one of the most restorative materials exhibiting a wide variety of novel properties such as low cost, low toxicity, and environmental friendly, along with high chemical stability in the pH range. Moreover, the bandgap of WO_3 (2.5–2.8 eV) is narrower than those of TiO_2 (3.2 eV for anatase) and ZnO (3.3 eV), which is suitable for reducing the electron/hole recombination. In addition, the hole diffusion length of WO_3 (~150 nm) is also longer than that of TiO_2 (~10 nm) [14]. Tungsten trioxide exhibits a cubic perovskite-like structure based on the corner-sharing of regular octahedra with the oxygen atoms at the corner and the tungsten atoms at the centre of each octahedron. Stoichiometric tungsten trioxide also shows structural polymorphism, and phase changes occur when the material is heated

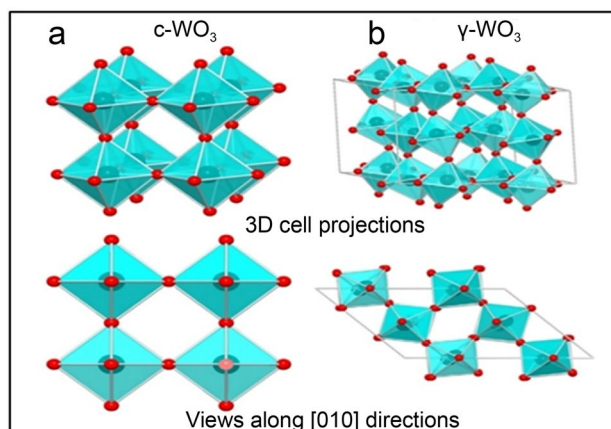


Fig. 2 Cell projections together with the views along the [010] crystallographic direction for **a** ideal cubic structure (c-WO_3) **b** the room temperature stable phase (monoclinic, $\gamma\text{-WO}_3$); the small red spheres represent the oxygen atoms and the large black spheres inside the polyhedra represent the tungsten atoms from Ref. [15]. Reproduced with permission from Ref. [15]. Copyright 2019, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

or cooled at different temperatures (Fig. 2) [15]. It is tetragonal at temperatures above 740 °C, orthorhombic from 330 to 740 °C, monoclinic from 17 to 330 °C, and triclinic from –50 to 17 °C [16]. The first application of WO_3 discovered in the 1920s is pH sensitivity [17]. Deb [18] showed that these WO_3 materials were more suitable for electrochromic applications. After a few years, in 1976, WO_3 was used for the photoanode in the water-splitting photoelectrochemical cell application [19, 20]. WO_3 has also been studied for gas sensor applications [21]. These tungsten trioxides can be used in various applications, i.e., surfactants, pesticides, dyes, phenols, chloro compounds, nitrogen-containing compounds, as an antimicrobial agent [22], lithium-ion batteries [23], supercapacitors [24], and smart windows [25]. Nowadays, it is studied as a photocatalyst [26] for pollutant reduction (reducing air pollutant gasses emitted during combustion processes to reduce environmental pollution and preserve the ecosystem) [27–29]. Platinum-activated tungsten trioxide was used to detect hydrogen gas [30] and electrode materials in the fuel cells [31]. Tungsten oxide has been proven to be a promising material exhibits different morphologies i.e. nanorods [21], nanowires [21], nanosheets [32], nano flakes [32], nanobelts [33], nanofibers [34] and nanoflowers [35]. Different methods have been used for the synthesis of tungsten oxide, such as chemical vapor deposition [36], thermal vapor deposition [37], heating tungsten filament/wire in a vacuum or atmosphere [38], anodization and electrodeposition [39], hydrothermal and solvothermal techniques [24, 32].

Even though WO_3 has promising features, further improvements are still required to make it efficient for

different applications. Pure WO_3 is not efficiently photo-activated due to the high electron–hole recombination rate and difficulty reducing oxygen [40]. Its slow switching speed does not meet the requirement of the display device and increases energy consumption [41]. To overcome this drawback, WO_3 -based nanocomposites such as WO_3 /carbon nanotubes, WO_3/Ag , WO_3/TiO_2 , and WO_3 graphene-based nanocomposites are expected to improve the electrochromic performance [9]. Different approaches can be used to generate different morphologies such as mesoporous and vertically aligned nanostructure arrays (nanowire, nanorods, and plate-like) to improve light harvesting and provide an effective channel for the directional transport of electrons. At the same time, doping WO_3 with metals or nonmetals to improve performance is an excellent way to tune the bandgap energy [14].

It has been observed that the composite of graphene with WO_3 nanomaterials is more suitable for various applications (Fig. 3), reflected by the rise of publications summing up to hundreds of citations over the ten years (Fig. 4). The composites can be prepared by hydrothermal method [7], pulsed laser ablation [8], electrochemical deposition [9], spray pyrolysis [10], microwave irradiation [11], and sol–gel method [12]. WO_3 /graphene-based nanocomposites are superior to other nanomaterials because when WO_3 is anchored on rGO, the rebuilding of the π – π conjugate in the WO_3 /rGO nanocomposites occurs. These alter the state density in the composite, leading to the shifting of the conduction band down and reducing the bandgap energy [12], which effectively reduces the electron/hole recombination and thereby improves the photodegradation performance of the composite [42]. It is also revealed that the bandgap of WO_3 decreased from 2.54 to 2.45 eV with the addition of rGO, suitable for water splitting and photocurrent generation applications [43]. A GO/ WO_3 and WO_3 /rGO hybrids also

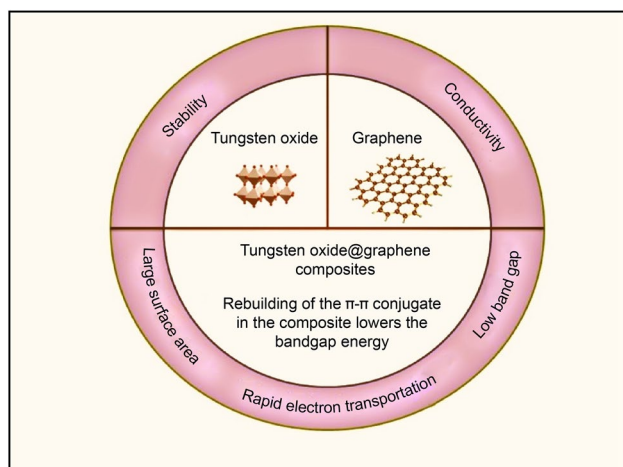


Fig. 3 A short view of WO_3 , graphene and WO_3 /graphene based nanocomposites with their advantages

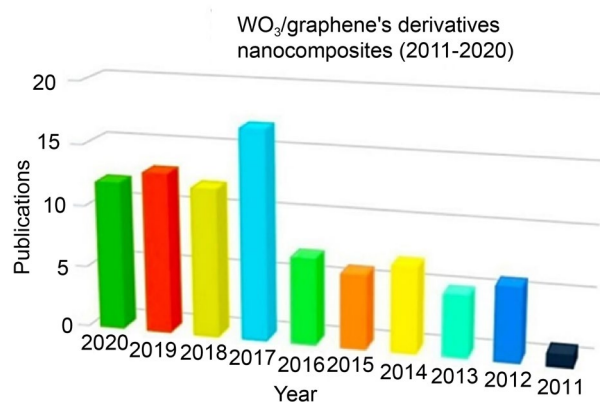


Fig. 4 Significant increase of WO_3 /graphene composite's publications in last ten years

offer rapid electron transportation, higher available superficial area offering more active sites leading to an increase in specific capacitances with extraordinary long cycle life [44, 45].

On the other hand, WO_3 /rGO composite film, WO_3 joined onto the outside of rGO sheets through C–W and C–O–W bonds. The C–O–W bond development makes the W=O bond vulnerable in WO_3 [12]. Furthermore, compared with GO, the characteristic absorption bands of C–O (alkoxy) stretching vibration weaken distinctly in the spectrum of the composites, indicating the decrease of oxygen-containing functional groups after hybridization [7]. Raman shift due to the tensile stress at the interface of WO_3 and rGO provides reliable evidence of charge transfer between the graphene sheets and the WO_3 in the WO_3 /graphene composite, suggesting a new bonding between the Graphene and WO_3 as shown in Fig. 5 [46].

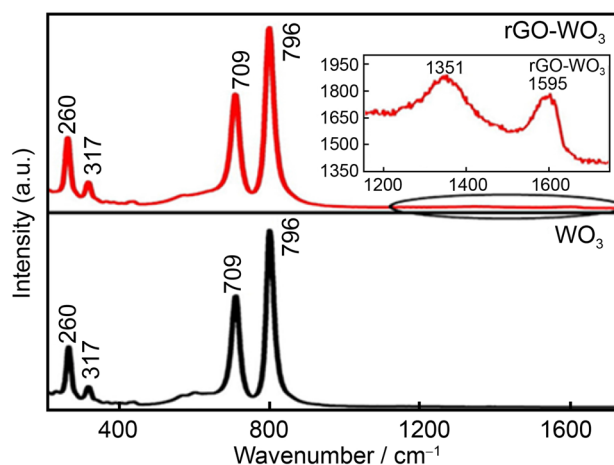


Fig. 5 Raman spectra of WO_3 and the rGO composite from Ref. [46]. Reproduced with permission from Ref. [46]. Copyright 2018, Elsevier

Nisa et al. [47] published a review article that summarizes the 5-year applications of GO/rGO-based tungsten oxide nanocomposites in energy storage (supercapacitors and batteries), gas sensor devices, electrochromism, and photocatalyst. However, they have engrossed only five applications. Still, the present review summarizes the 10-year applications of graphene-based tungsten oxide nanocomposites in photocatalysis and photo electrocatalysis, energy storage, medical, electrochromism, and energy conversion [47]. In this review, the noteworthy endeavors and exceptional turning points accomplished utilizing heterojunction of WO_3 and derivatives of graphene for different applications. The current article will act as a bridge, filling a research gap that will stimulate new ideas for further improvement in graphene-based tungsten oxide nanocomposites. Conclusively, the scope of future research work to design the ternary composite with high efficiency utilizing WO_3 and graphene is also explored.

2 Synthesis methods

Graphene@tungsten oxide-based composite can be prepared by both physical and chemical methods. We have discussed most prevalent deposition methods, including both chemical and physical methods (Fig. 6).

2.1 Chemical methods

2.1.1 Solvothermal and hydrothermal methods

The hydrothermal process is one of the most familiar and extensively used methods to produce nanostructured materials in liquids at high temperatures and pressure. In this method, nanostructured materials are attained through a heterogeneous reaction carried out in an aqueous medium [48].

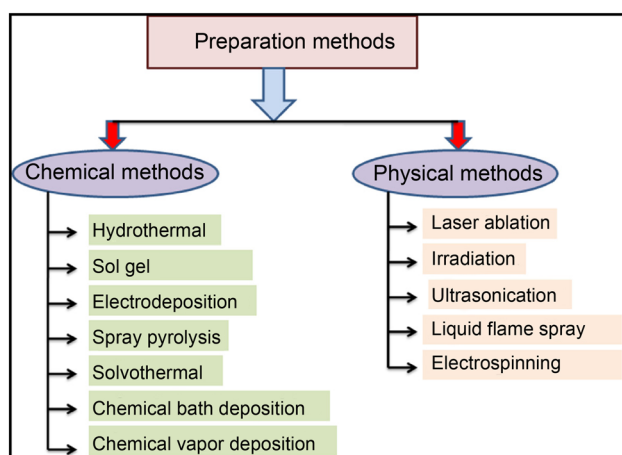


Fig. 6 Preparation methods of WO_3 /graphene based nanocomposites

In comparison, solvothermal methods are generally carried out in a non-aqueous medium. Hydrothermal and solvothermal methods are usually carried out in closed systems (autoclave) [49]. The autoclave is hermetically sealed, and at the lower side, heating is applied, resulting in a temperature gradient between the lower and the upper ends. Consequently, due to the supersaturation of the solute in the cooler zone, nucleation, followed by precipitation and growth of the compounds, occurs [50]. Different morphologies, sizes, or phases can be achieved by changing the solution pH value, reaction temperature, and solute species' concentration or the additives' types [51]. The microwave-assisted hydrothermal method has recently received significant attention for engineering nanomaterials, combining the merits of both hydrothermal and microwave methods [15].

2.1.2 Sol–gel method

Sol–gel coating technology has received considerable interest in the last 20 years due to its low production costs and broad coating area capability [15]. These characteristics make sol–gel an appealing technique for commercial applications [52]. This process is known as a sol–gel method because the liquid precursor is turned into a sol during the synthesis of the metal-oxide nanoparticles. The sol is eventually converted into a network structure known as a gel [53]. The concept is based on producing an oxide network by polymerization reactions of molecular precursors. [52] Metal alkoxides are common precursors for the sol–gel process of producing nanomaterials. The sol–gel method of nanoparticle production can be performed in multiple steps. In the first step, the metal oxide is hydrolyzed in water or with the help of alcohol to generate a sol. Condensation occurs next, resulting in an increase in solvent viscosity and the formation of porous structures that are left to age. Hydroxo- ($\text{M}-\text{OHM}$) or oxo- ($\text{M}-\text{O}-\text{M}$) bridges develop during the condensation or polycondensation process, resulting in metal-hydroxo- or metal-oxo-polymer formation in solution [54]. Polycondensation persists during the aging process, resulting in structure, characteristics, and porosity changes. Porosity reduces with aging and the space between colloidal particles increases. Following the aging process, the gel is dried, and water and organic solvents are removed. Finally, calcination is used to produce nanoparticles [55].

2.1.3 Chemical vapor deposition (CVD)

Chemical vapor deposition (CVD) is efficient for producing high-quality solid thin films [55]. In this method, a thin coating is made on the surface by the chemical reaction of vapor-phase precursors [56]. A precursor is ideal for CVD if it has sufficient volatility, high chemical purity, strong evaporation stability, cheap cost, non-toxic and long life. Moreover, the

decomposition of the ideal precursor should not produce leftover contaminants [55]. In this methodology, the catalyst can tailor the morphology and type of nanomaterials [56]. In simple terms, CVD is an excellent method for producing two-dimensional nanomaterials [57].

2.1.4 Electrochemical deposition

Electrodeposition is a well-established coating process that uses an applied voltage in electrolytic cells to swiftly assemble ions, polymers, and colloids in significantly less time [58]. It is a very appealing coating technology because of its low cost, significant area coating capability, and fine thickness control. The setup consists of an electrochemical cell filled with the electrolyte with the metal species to be deposited and two (or three) electrodes: the cathode, which is a conducting substrate onto which the film grows, the anode (generally Pt), and a reference electrode if a three-electrode arrangement is used. A direct current is applied to the circuit, causing the metal species in the solution to be driven to the cathode, where they are reduced and electrodeposited to create the film [59]. The resulting film microstructure can be accurately adjusted by changing the process parameters and electrolyte composition [55].

2.2 Physical methods

2.2.1 Laser ablation

Laser ablation synthesis generates nanoparticles by striking the target material with an intense laser beam. Because of the high energy of the laser irradiation, the source material or precursor vaporizes during the laser ablation process, resulting in nanoparticle production. Because no stabilizing agents or other chemicals are required, using laser ablation to generate noble metal nanoparticles can be called a green process [60]. This process may create a wide variety of nanomaterials, including metal nanoparticles [61], carbon nanomaterials [48, 62], oxide composites [51], and ceramics [63]. Pulsed laser ablation in liquids to create monodisperse colloidal nanoparticle solutions without the need for surfactants or ligands is an interesting new method. The fluence can be adjusted to tailor nanoparticle parameters like average size and distribution [55].

2.2.2 Pulsed laser deposition (PLD)

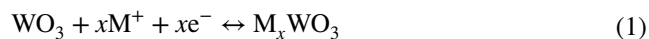
Pulsed laser deposition (PLD) is one of the best coating methods for researching the fundamental characteristics of materials [64]. It is widely employed due to its ability to produce films that consistently reproduce the intended stoichiometry and its adaptability in tuning from amorphous to crystalline structures. The principle is based on ablating a

ceramic target surface with an excimer laser, resulting in plasma formation. The film is deposited onto a substrate positioned in front of a plasma plume containing atom clusters and molten droplets of material [55, 65].

3 WO₃/rGO composite nanomaterials for different applications

3.1 Electrochromic applications

Smart windows have recently received a lot of attention since energy security needs effective energy solutions to offset the use of traditional energy sources [9]. Tungsten trioxide has emerged as a promising electrochromic (EC) material that can switch between Transparent and blue (Fig. 7) with a relatively fast response time and high coloration efficiency compared with other electrochromic materials [9]. Electrochromism in WO₃ can be achieved through the injection–extraction of electrons and H⁺ or Li⁺ cations, resulting in a color change [66]. Electrochromism in WO₃ can be achieved through the injection–extraction of electrons and H⁺ or Li⁺ cations, resulting in a color change [67]. Electrochromism in WO₃ can be given by simple reaction (Eq. 1).



With $\text{M}^+ = \text{H}^+, \text{Li}^+, \text{Na}^+, \text{or } \text{K}^+$, whereas e^- are denoting electrons. Thus WO₃, which is transparent in oxidation state, incorporates electrons and charge-balancing ions and can be reduced to a material with radically different properties [67].

In practice, WO₃-based EC devices have a number of drawbacks, including a long switching time and low stability [68]. With the addition of rGO, the bandgap of WO₃ was reduced from 2.54 to 2.45 eV, offers more rapid electron

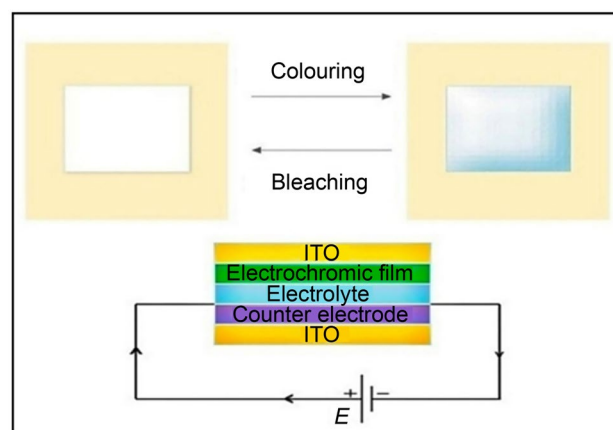


Fig. 7 Phenomenon of electrochromism and structure of an electrochromic device

transportation resulting in short switching time and longer cycle life [44, 45].

A great deal of effort has been presented forward in developing WO_3/rGO nanocomposites based on electrochromism. The electrochromic behavior of WO_3/rGO nanocomposites synthesized by the electrodeposition method shows higher colouration efficiency, long cycle lifetime, and fast switching time than the WO_3 films alone. It can be concluded that the rGO in the composite film offers rapid electron transportation, a higher available superficial area offering more activity, shortening the path of ion diffusion, and allowing the electrolyte to penetrate easily through the membrane [9]. The sol–gel spin coating technique was utilized to fabricate WO_3/rGO nanocomposites film in 2016. Experimental results showed that the developed nanocomposites demonstrated shorter colouration–bleaching times, higher colouration efficiency, more extensive optical modulatory range, and better cyclic stability compared with WO_3 films, which are attributed to faster Li^+ ion diffusion and electron transfer rate [12]. The thin film of WO_3/rGO nanocomposites fabricated by the sol–gel method shows a better switching speed with high reversibility and contrasted with the new WO_3 films inferring ethylene glycol (EG) plays an essential role in the reduction of GO. Furthermore, the prepared porous structure is more suitable for enhancing the insertion/extraction of ions or electrons in the electrodes [69].

The performance of the WO_3/rGO nanocomposites synthesized by spray pyrolysis on ITO glass substrate was reported in 2019 [70]. It is revealed that WO_3/rGO film was reversible in nature in -2.1V potential and had an effect of increasing the light transmission coefficient at a positive $+2\text{V}$ potential and restoring in the absence of potential. The porous rGO/ WO_3 composite films were prepared using the hydrothermal method. The higher colouration efficiency of WO_3/rGO film than WO_3 , rGO films is observed due to the close contact between the WO_3 nanorods (NRs) and rGO, which is very important for enhancing EC performance of WO_3/rGO composites [67]. Similarly, the optical and electromagnetic properties of the WO_3/rGO EC nanocomposites were obtained by mechanical spraying of a water-based dispersed solution with WO_3/GO particles. It has been demonstrated that annealing in an inert argon atmosphere results in forming an electrically conductive phase of rGO and crystalline WO_3 [71]. Thus, the performance of electrochromic devices has been improved by doping rGO in WO_3 . The comparison between the results of these WO_3/GO composites prepared under different experimental conditions is tabulated in Table 1. It indicates the improved electrochromic performance of $\text{WO}_3/\text{graphene}$ -based nanocomposites.

Nevertheless, electrochromic devices (ECDs) do not change color automatically and require an external power source to achieve electrochromism [70] which is not easy to carry and is associated with energy consumption. Solar

cells can convert solar energy into electric energy and are environmental friendly. By connecting solar cells with ECDs for storage [72], photo-electrochromism can be realized to effectively adjust visible and near-infrared light transmittance without any external power supply providing zero energy consumption and converting solar energy into electrical energy.

3.2 As lithium-ion batteries and supercapacitors

In the recent era, there are urge power and energy density demand from mobile electronics and electric vehicles, and Li-ion and supercapacitors technology are being tried to fulfill this demand. The various metal oxide electrodes are used for supercapacitors and battery applications [73]. A Li-ion battery is constructed by connecting primary Li-ion cells in parallel (to increase current), series (to increase voltage), or combined configurations. Typically, a primary Li-ion cell consists of a cathode (positive electrode) and an anode (negative electrode) which are contacted by an electrolyte containing lithium ions. The two electrodes are connected externally to an external electrical supply during the charging process. The electrons are forced to be released at the cathode and move externally to the anode. Simultaneously the lithium ions move in the same direction, but internally, from cathode to anode via the electrolyte. In this way, the external energy is electrochemically stored in the battery in the form of chemical energy in the anode and cathode materials with different chemical potentials. The opposite occurs during the discharging process: electrons move from the anode to the cathode through the external load to do the work, and Li ions move from the anode to the cathode in the electrolyte. This is also known as the “shuttle chair” mechanism, where the Li-ions shuttle between the anode and cathodes during the charge and discharge cycle [74].

Supercapacitors primarily store electrical energy by forming a double-layer capacitor structure at the electrodes and the electrolyte interface. The reversible faradic reaction on the electrode surface contributes to total capacitance and electrostatic charge transfer results in high recyclability [75]. The transition metal oxide WO_3 has gotten much attention as a potential electrode material for lithium-ion batteries [23] and supercapacitors [24]. It has a high intrinsic density ($> 7 \text{ g}\cdot\text{cm}^{-3}$), a prominent theoretical capacity ($700 \text{ mAh}\cdot\text{g}^{-1}$), a high energy density, and excellent cycling stability [76–78]. However, pure WO_3 has some disadvantages, such as poor electrical conductivity, cyclic instability, and a large irreversible capacity [79]. As a result, it must be combined with a highly conductive carbonaceous material, such as carbon nanotubes [80], graphene [81], or reduced graphene oxide [82] to improve the overall performance of WO_3 . According to the theoretical explanations for improved capacitance in hybrid structures, it is proposed that the

Table 1 Electrochromic performances of WO₃/graphene-based nanocomposites reported in the literature

Material	Experimental condition	Electrochromic response		References
rGO/WO ₃	Precursor: sodium tungstate, propylene glycol, HCl G.O. was prepared via the modified hummers method Temp: 100 °C Time: 1 h	Sample	Response time (s)	[66]
			T_c	T_b
		WO ₃	2.0	2.13
		rGO	1.5	0.6
		WO ₃ /rGO	4.7	4.5
		sample	C.E (cm ² ·C ⁻¹)	
		WO ₃	122.2	
		WO ₃ /rGO	181.5	
WO ₃ /graphene	Precursor: tungstate powder hydrogen peroxide, ethanol Graphene oxide was prepared via the modified hummers method	WO ₃ /rGO film exhibits optical modulation 58.8% at 633 nm		
		sample	Response time (s)	[9]
			T_c	T_b
		WO ₃	10.1	15.7
		WO ₃ /rGO	6.8	9.1
		sample	C.E (cm ² ·C ⁻¹)	
		WO ₃	96.1	
		WO ₃ /rGO	66.7	
WO ₃ /rGO	Precursor: W powder, H ₂ O ₂ , glacial acetic acid and ethanol Time: 70 h Temp: room temp G.O. was prepared via modified hummers method	WO ₃	The contrast remains 27% in the initial state after 300 cycles	
		WO ₃ /rGO	The contrast remains 85% in the initial state after 300 cycles	
		sample	Response time (s)	[69]
			T_c	T_b
		WO ₃	15.5	16.5
		WO ₃ /rGO	9.0	4.5
		Sample	Reversibility (%)	
		WO ₃	87.8	
WO ₃ /rGO	Precursor: Natural graphite powder, KMnO ₄ , H ₂ SO ₄ , Wl ₆ , NaNO ₃ , anhydrous ethanol Time: about 4 d Temp: 50 °C for 24 h GO was prepared via modified hummers method	WO ₃ /rGO	95.4	
		Sample	Ion storage capacity (mC·cm ⁻²)	
		WO ₃	11.1	
		WO ₃ /rGO	21.8	
		sample	Diffusion coefficient (m ² ·s ⁻¹)	
		WO ₃	1.1×10^{-10}	
		WO ₃ /rGO	4.9×10^{-10}	
		Sample	Response time(s)	[12]
WO ₃ /rGO			T_c	T_b
		WO ₃	14.3	10.8
		WO ₃ /rGO	9.5	7.6
		Sample	C.E (cm ² ·C ⁻¹)	
		WO ₃	46.7	
		WO ₃ /rGO	75.3	
		Sample	Optical modulation (%)	
		WO ₃	53.8	
		WO ₃ /rGO	59.6	

T_c , T_b , Response time for colored and bleached state; C.E, coloration efficiency

interaction between the WO₃ and rGO layer involves Van der Waals forces and chemical bonding and charge transfer. The C 2p orbital of rGO is hybridized with the d orbital of W and the 2p orbital of O, resulting in additional electronic

states near the Fermi level and an increase in capacitance in hybridized WO₃ and rGO compared to pristine WO₃ [79].

In 2013, WO₃ nanowires/graphene nanocomposites were synthesized via a facile hydrothermal method and

evaluated as an anode for lithium-ion batteries. Improved electrochemical performance could be ascribed to incorporating highly conductive graphene into a three-dimensional (3D) nanostructure that mitigates the capacity degradation at high current densities of electrodes for lithium-ion batteries [83]. Sandwich structures have attracted tremendous attention owing to their unique properties such as enhanced energy absorption, incredible stiffness to weight ratios, excellent ballistic resistance performance, and good thermal and acoustic isolation properties. It is a multipurpose novel structure made up of two or more individual components with different properties that, when combined, form a high-performance material [84]. Several advantages can be gained from the sandwich structure of WO_3 , including more active sites, larger specific reaction surface area, effective absorption of light, and separation of the photogenerated electrons and holes that may be used in different applications. Additionally, it provides low charge-transfer resistance owing to improved photoanode current density. Creating a sandwich structure with superior crystallinity can effectively facilitate the separation of photogenerated electron-hole pairs and suppress the formation of peroxo-species, making for different applications [85]. Hierarchical sandwich-type tungsten trioxide nanoplatelets/graphene-based nanocomposites as anode materials were reported in 2016 [86]. It is found that sandwich-type architecture plays a vital role in endowing high electronic conductivity, accommodating considerable volume variation, and achieving abundant ion transport paths, thus enabling superior lithium-ion storage performance [86]. Novel 2D mesoporous $\text{WO}_3\text{-x}$ /graphene sheets using mesoporous silica with graphene oxide as an electrode for Li-ion battery were fabricated by template approach. The experimental results revealed that the mesoporous $\text{WO}_3\text{-x}$ closely adheres to the graphene nanosheets and has vertical channels that can ease electron and Li^+ diffusion moreover enhance mechanical stability during cycling [87]. The comparative supercapacitive property study of $\text{WO}_3/\text{WO}_3\cdot\text{H}_2\text{O}$ mixture electrode and graphene nanosheets/ WO_3 was also investigated. It is exposed that graphene nanosheets/ WO_3 exhibited high improved specific capacitance due to the rapid electron transportation and higher available superficial area offering more active sites [88]. GO/WO_3 hybrids show remarkable improvement in specific capacitance ($580 \text{ F}\cdot\text{g}^{-1}$) with excellent long cycle life due to the increase in electro-conductivity, facilitating efficient charge transport and promoting electrolyte diffusion [44] (Fig. 8).

It is revealed that graphene- WO_3 nanocomposites with an optimized weight ratio have shown excellent electrochemical performance with high specific capacitance. Morphological characterization shows the distribution of WO_3 nanowires on the graphene sheets, preventing stacking of graphene sheets and thus exhibiting a large surface area. It can be concluded that the large surface area of graphene- WO_3 nanowires

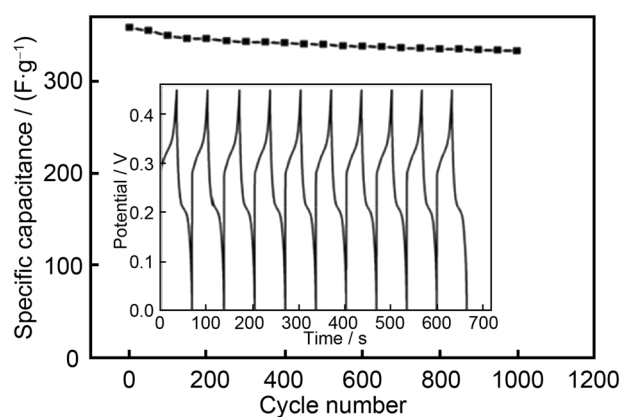


Fig. 8 Cyclic performance of Gr- WO_3 hybrid electrode at 5 A g^{-1} in 2 mol L^{-1} KOH electrolyte from Ref. [44]. Reproduced with permission from Ref. [44]. Copyright 2016, the Royal Society of Chemistry

nanocomposites facilitates higher specific capacitance making it an efficient active electrode material for the fabrication of supercapacitor devices [89]. One-pot and fast route for developing WO_3/rGO nanocomposites as a supercapacitor was reported for the first time. It has been discovered that doping WO_3 with rGO results in a material with synergic properties such as faster electron transport, higher available superficial, and higher specific capacitance [79]. Promising supercapacitor electrode material (WO_3/rGO hybrids) synthesized by a simplistic one-pot hydrothermal synthesis. Experimental findings reveal that the doping of rGO in WO_3 leads to more rapid electron transportation, higher available superficial, and an improved specific capacitance [45].

In 2020, a unique approach for creating a WO_3/rGO heterojunction using poly (ionic liquid) (PIL) as a linker was reported [90]. The findings of the experiments demonstrated that PIL acts as a good contact between the WO_3/rGO and considerably improves the supercapacitive performance of composite electrodes [90]. The specific capacitance of $\text{WO}_3\text{-rGO}$ composites was increased by intimately coating WO_3 nanoflowers with rGO via electrostatic interaction between positively charged modified WO_3 and negatively charged GO. The enhancement in specific capacitance and excellent cyclic stability owing to sufficient interfacial contact and the synergetic impact of pure WO_3 and graphene. The experiments showed that the $\text{WO}_3\text{-rGO}$ composite is a promising electrode material for high-performance supercapacitors [91]. WO_3 nanowires directly grown on graphene sheets have been fabricated using a seed-mediated hydrothermal method. The experimental results show that the precoated moonseeds and graphene sheets on graphite electrodes provide more reactive centres for nucleating and forming uniform WO_3 nanowires. Over a voltage range of 1.6 V, a high-performance electrochemical supercapacitor assembled with WO_3 nanowires as a negative electrode and

Polyaniline (PANI)/MnO₂ as positive electrodes displays a high volumetric capacitance of 2.5 F·cm⁻³, indicating that WO₃ nanowires on graphene sheets as negative electrode have a lot of potential for energy storage devices [92]. The morphology and heterojunction were found to be quite beneficial in improving the overall performance of the composite [93]. It is described that one-dimensional (1D) nanomaterials shorten free electron scattering and have a larger surface-to-volume ratio than nanoparticles [94, 95]. It has been perceived that 1-D WO₃ nanorods provide:

- More potential electron routes between WO₃ and graphene sheets.
- Resulting in more reactive spots on the surface of both graphene and WO₃ nanorods [96, 97].
- Making 1D/2D heterojunction suitable candidate for various applications.

In 2D/2D heterojunctions, due to the difference in Fermi levels between 2D WO₃ and 2D rGO sheets, it was found that the introduction of 2D rGO resulted in a positive downgrade of conduction band (CB) and valence band (VB) from bare 2D WO₃ to 0.22 and 2.6 eV, respectively. In addition, when the 2D/2D rGO/WO₃ heterojunctions were exposed to simulated sunlight, the excited electron jumped to the

CB of 2D WO₃. The excited electron rapidly moved to the rGO, efficiently separating the photogenerated electron-hole pairs. As a result, the light absorption capacity and utilization of photoelectron-hole pairs formed in 2D WO₃ were significantly improved by 2D rGO sheets [98]. Furthermore, 2D WO₃ on rGO facilitated the surface interaction between the mixed sections, which enabled the transfer of charge carriers from the 2D sheet of WO₃ to rGO causing reduced recombination of electrons and holes [47]. Hence, 2D WO₃ nanoplates decorated on graphene sheets exhibit superior electrochemical properties as composites possess higher surface area, offering more active sites, ensuring enhanced electrochemical properties during the electrochemical reaction [11].

In 2019, it was demonstrated that optical characterization of nanocomposites fabricated by Pulsed laser ablation exhibited better visible light absorption and less photogenerated charge recombination than pure WO₃. It can be concluded that the attractive electrochemical and structural characteristics, along with the high electrical conductivity and porosity of rGO, enable the enhancement of the capacitance of the supercapacitor compared to pure WO₃ [8]. Morphology can provide a direct pathway for optically generated charges with reduced grain boundaries, leading to superior charge transport properties [93]. Flower-like WO₃-H₂O/reduced

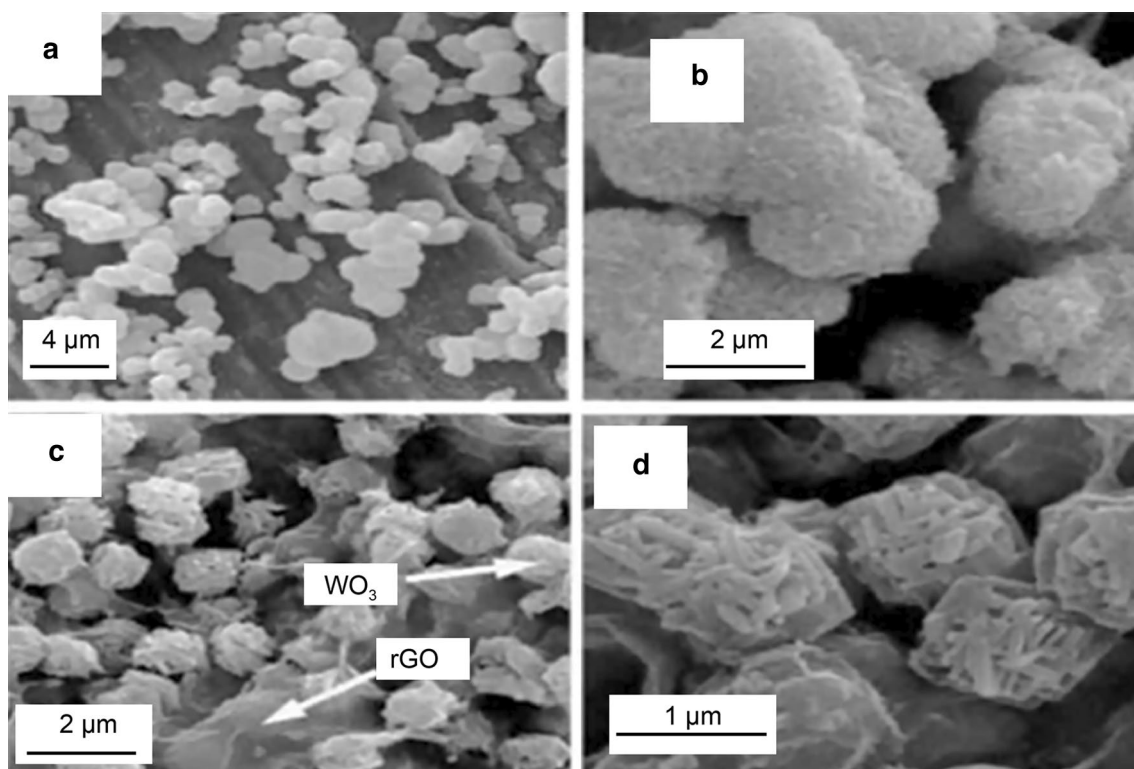


Fig. 9 SEM images of bare WO₃ (a, b) and WO₃-H₂O/rGO (c, d) from Ref. [24]. Reproduced with permission from Ref. [24]. Copyright 2015, Elsevier

graphene oxide composite with excellent electrochemical stability (Fig. 9) has been successfully synthesized through a mild hydrothermal approach. It has been discovered that the orthorhombic phase of $\text{WO}_3\text{-H}_2\text{O}$ in composite has a flower-like hierarchical design that is equally dispersed and anchored on the vast and flexible reduced graphene oxide sheets, with homogeneous sizes and morphologies. It can be concluded that the increased electrochemical capabilities are thought to be due to the unique hierarchical $\text{WO}_3\text{-H}_2\text{O}$ microstructures and the synergistic impact resulting from the intimate interaction of $\text{WO}_3\text{-H}_2\text{O}$ with rGO [24]. Spray pyrolysis was used to synthesize $\text{WO}_3\text{-rGO}$ composite powders, employing a colloidal stable graphene oxide solution with ammonium tungstate. The shape and crystal structure of WO_3 powders were altered by GO nanosheets [11]. WO_3 as an electrode material for supercapacitors has always suffered from low capacitance and poor rate capability [99]. The hydrothermal approach was successfully used to synthesize WO_3 NRs/rGO composites with various rGO ratios. Because of the combined action of the double layer and pseudocapacitors, the WO_3 NRs/rGO composite demonstrates high specific capacity, rate performance, and cycling stability when compared to pure WO_3 electrodes [99]. The comparison between the results of the composite of WO_3 /graphene and its derivative prepared under different experimental conditions is represented in Table 2.

3.3 Gas sensing applications

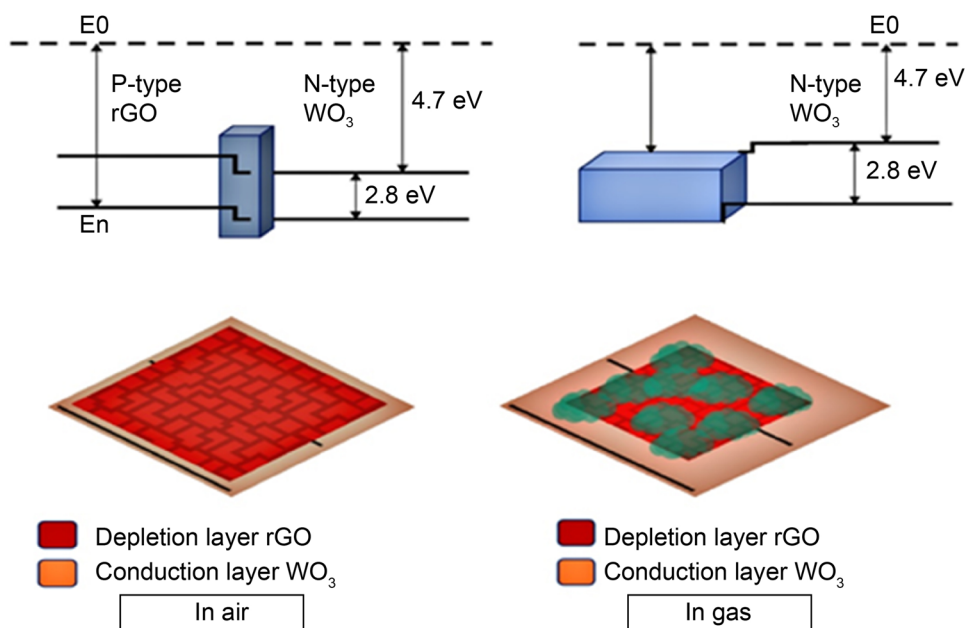
The composite $\text{WO}_3\text{/rGO}$ thin-film sensors work on the change in electrical conductivity or resistivity on exposure to a target gas. The change of resistance of the thin film depends upon the type of majority carriers in the semiconducting materials and the nature of gas molecules [100, 101]. The WO_3 is an n-type semiconductor metal oxide with highly dominant electron charge carriers. The gas-sensing mechanism of rGO is generally p-type semiconducting behavior and dominated by positive charge carriers, and the surface serves as electron donors, and exposure to an oxidizing gas ought to decrease their resistance [102]. When the n-type WO_3 conduction layer and p-type depletion layer of rGO transfer the electron from WO_3 into rGO, the Fermi level of WO_3 lines up with that of rGO, causing a large interfacial barrier. The barrier construction further speeds up the initial resistance of the $\text{WO}_3\text{/rGO}$ hybrids, causing an increased response. Upon exposure to the oxidative gas, the number of electrons becomes fewer while the barrier becomes stronger with rising in the work function of WO_3 . Upon exposure to the reducing gas, the electrons previously trapped by oxygen atoms and rGO are instantaneously released back into the WO_3 nanostructure, and then electron diffusion occurs between $\text{WO}_3\text{/rGO}$ sheets. This narrows the region barrier and hence significantly reduces the resistance of the $\text{WO}_3\text{/rGO}$ system shown in Fig. 10 [103].

According to space-charge layer model [100], the gas-sensing response of WO_3 involves formation of a charge depletion layer in the near-surface region of each grain due

Table 2 Electrochemical responses of WO_3 /graphene-based nanocomposites reported in the literature

Sr No	Material	Capacitance	Life cycle	References
1	$\text{WO}_3\text{-rGO}$	$577 \text{ F}\cdot\text{g}^{-1}$ at $1.0 \text{ A}\cdot\text{g}^{-1}$	—	[8]
2	$\text{WO}_3\text{-rGO}$	$546 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$ at $100 \text{ mA}\cdot\text{g}^{-1}$	300 cycles	[10]
3	$\text{WO}_3\text{/G}$	$761 \text{ F}\cdot\text{g}^{-1}$ at $1.0 \text{ A}\cdot\text{g}^{-1}$	99.2% over 5000 cycles	[11]
4	$\text{WO}_3\text{-H}_2\text{O/rGO}$	$244 \text{ F}\cdot\text{g}^{-1}$ at $1 \text{ A}\cdot\text{g}^{-1}$	20% over 900 cycles	[24]
5	Graphene- WO_3	$580 \text{ F}\cdot\text{g}^{-1}$ at $1 \text{ A}\cdot\text{g}^{-1}$	92% over 1000 cycles	[44]
6	rGO/ WO_3	$287 \text{ F}\cdot\text{g}^{-1}$ at $0.3 \text{ A}\cdot\text{g}^{-1}$	70% over 3000 cycles	[45]
7	$\text{WO}_3\text{NRs/rGO}$	$801.6 \text{ F}\cdot\text{g}^{-1}$ at $4 \text{ A}\cdot\text{g}^{-1}$	75.7% over 5000 cycles	[79]
8	3D WO_3 nanowires/graphene	$656 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$ at $100 \text{ mA}\cdot\text{g}^{-1}$	22.7% over 100 cycles	[83]
9	Tungsten trioxide nanoplates/graphene	$847 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$ at $72 \text{ mA}\cdot\text{g}^{-1}$	118.9% at $1800 \text{ mA}\cdot\text{g}^{-1}$ after 1000 cycles	[86]
10	2D mesoporous WO_{3-x} /graphene sheets	Charge $1451 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$	Discharge $761 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$ 100 cycles	[87]
11	Graphene nanosheets-tungsten oxide	$143.6 \text{ F}\cdot\text{g}^{-1}$ at $0.1 \text{ A}\cdot\text{g}^{-1}$	—	[88]
12	Graphene- WO_3 nanowire	$465 \text{ F}\cdot\text{g}^{-1}$ at $1.0 \text{ A}\cdot\text{g}^{-1}$	97.7% over 2000 cycles	[89]
13	$\text{WO}_3\text{/rGO}$	$316 \text{ F}\cdot\text{g}^{-1}$ at $1.0 \text{ A}\cdot\text{g}^{-1}$	84.3% over 2000 cycles	[90]
14	$\text{WO}_3\text{/rGO}$	$495 \text{ F}\cdot\text{g}^{-1}$ at $1 \text{ A}\cdot\text{g}^{-1}$	87.5% over 1000 cycles	[91]
15	WO_3 nanowires/graphene	$800 \text{ mF}\cdot\text{cm}^{-2}$ at $1 \text{ mA}\cdot\text{cm}^{-2}$	—	[92]
16	WO_3 NRs/rGO	$343 \text{ F}\cdot\text{g}^{-1}$ at $0.2 \text{ A}\cdot\text{g}^{-1}$	—	[99]

Fig. 10 Gas sensing mechanism of WO_3/rGO based Nanocomposite



to electron trapping on adsorbed oxygen species. Oxygen molecules from the gas will be adsorbed on the surface and the grain boundary of WO_3 , capturing the electron from the conduction bands to form the negative oxygen ions O_2^- via O^- to O_2^- , with the increase of the temperature. As a result, the optimum operating temperature with NO_2 gas is 100–120 °C [104], whereas Aniline gas is up to an optimal operating temperature of 80 °C [105]. Room temperature sensing of NH_3 is about 32–35 °C [106] and O_3 gas-operated temperature is 150 °C [107]. The optimized temperatures for Acetone gas were found to be 350 and 200 °C [108]. H_2 gas sensor made by coupling the proton-conducting GO membrane with WO_3 shows a good sensor response of more than 90 mV to 300 ppm H_2 in the air at room temperature [109]. The depletion layer of rGO on the surface of the WO_3 composite film becomes thin, and the sensor's electrical resistance decreases. Thus, according to the response formula ($S = R_a/R_g$, where R_a and R_g are the resistance in air and target gas, respectively), the sensor based on WO_3 is sensitive to gasses environments. The sensor may perform a high response; consequently, the reasons are: graphene absorbs more gas molecules due to its high specific surface area with higher electrical conductivity, which improves the conductivity of WO_3 -rGO and results in electrons quickly spreading to the surface of the semiconductor [108–110]. The electrochemical activity of WO_3/rGO in catechol molecule oxidation is due to the charge transfer from O at the 2p orbital of catechol to the W 4d orbital of WO_3 facilitates the catechol molecule oxidation [111].

The present review directs particular attention to some breakthrough development in gas sensors based on WO_3 /graphene, and its derivatives-based nanocomposites has

shown excellent sensing response to NH_3 , NO_2 , H_2S , HCl , and volatile compounds [7, 112–126]. Subsequently, some information from related works such as detection limit, sensing range, the response time (t_{res})/recovery time (t_{rec}), repeatability, and stability are likewise concisely and carefully posed and discussed. Table 3 summarizes recent studies on diverse WO_3 graphene-based nanocomposites with possible applications as gas sensors.

3.4 Photocatalytic application

Recently, photocatalysis has been an effective oxidation process that has attracted much attention because of its low energy requirements and ease in use in breaking down organic pollutants into mineral materials. The semiconductor-based heterogeneous photocatalysis is one of the most promising candidate materials for environmental degradation due to the availability of solar light [127]. Photocatalysis is a method of driving chemical reactions with the help of light energy. The light is absorbed by electrons, which cause excitation from the valence band to the conduction band, leaving positive holes in the valence band. Photogenerated electrons react with O_2 to form a superoxide radical, which reacts with the pollutant to produce H_2O . Meanwhile, photogenerated holes are either transferred to the adsorbed pollutant, causing rapid oxidation or transferred to adsorbed water molecules, generating hydroxyl radicals, which react with the pollutant to produce the respective oxidation products [128]. Semiconductor metallic oxide photocatalysis, which includes WO_3 in photochemical oxidation of organic compounds observed through reactive oxidants, helps surface loading of plasmonic metals, anchoring natural

Table 3 Gas sensing performance of WO₃/graphene-based nanocomposites reported in the literature

Material	Gas sensed	Response	References
WO ₃ /rGO hybrids	H ₂ S	The response and recovery times of the WO ₃ /rGO sensor are ~ 340 s and ~ 180 s to 500 ppb of H ₂ S	[103]
Graphene (GR)-wrapped WO ₃ nanospheres	NO ₂ gas	The response of GR–WO ₃ nanocomposites to 7, 14, 28, 42 and 56 ppm NO ₂ are 11.6%, 21.3%, 26.1%, 32.9%, and 40.8%, respectively	[104]
WO ₃ /graphene	Aniline	Sensitivity of the WO ₃ /graphene sensor was about 4.7 to 1 ppm of aniline and only 1.8 of pure WO ₃ . WO ₃ /graphene has a most superior sensitivity to aniline among the tested gasses with a 100 ppm concentration	[105]
rGO/WO ₃ hybrid	NH ₃ gas	At room temperature, the sensing response 4.35 with 13/20 s response/recovery time for 10 ppm NH ₃ gas. The sensing performance has been obtained for 150 °C. For 10 ppm and 100 ppm NH ₃ gas, sensor response 10.89 and 27.7 with 11/17 and 7/19 s response/recovery time	[106]
WO ₃ -rGO nanocomposites	O ₃ gas	The response of sensor operated at 150 °C is almost proportional to the concentration ranging from 0.5 to 10 ppm Higher content of rGO the films gives rise to sensing response and the WO ₃ –3.0 wt.% rGO sensor has almost 4 times sensing response better than the pure WO ₃ sensor	[107]
WO ₃ nanolamella/rGO	Acetone	The optimized temperatures for which WO ₃ and rGO/WO ₃ show maximum response found to be 350 and 200 °C The variation in sensing response with different GO contents (0.5 wt.%, 1.0 wt.%, 2.0 wt.% and 4.0 wt.%) rGO/WO ₃ nanocomposite for 20 ppm of acetone at 200 °C is observed	[108]
WO ₃ /GCE/graphene	Ranolazine (RZ) detection	Two linear ranges for RZ determination were obtained between 0.2–1.4 µmol·L ⁻¹ and 1.4–14 µmol·L ⁻¹ with the low detection limit of 0.13 µmol·L ⁻¹ , satisfactory recovery (99.8% to 100.2%) with pharmaceutical samples	[112]
Graphene–WO ₃	NO ₂ gas	The sensor response for WO ₃ and 0.2 wt.% graphene–WO ₃ nanocomposite sensors at different temperatures. It is cleared that the response as well as the recovery is very fast in comparison to WO ₃ -multi walled carbon nanotubes (MWNTs) sensors	[113]
WO ₃ nanorods/graphene	NO ₂ gas sensing	The response of WO ₃ nanorods/3.5 wt.% graphene nanocomposites to NO ₂ was up to 25 times higher than that of pure WO ₃ nanorods	[114]
rGO/WO ₃	NO ₂ gas	The response and recovery times were 9 and 18 min, respectively. Ambient humidity significantly affected the NO ₂ gas sensor	[115]
WO ₃ -rGO porous nanocomposite (hao)	NO ₂	The original GO sample has a weak sensing response to NO ₂ in the temperature range 20–220 °C. Nevertheless, the composite WO ₃ -rGO has higher response toward NO ₂	[116]
WO ₃ /Graphene	NO _x detection	Graphene/WO ₃ nanorods exhibited a minimum response time of 29 s and recovery time of 24 s at room temperature for 10 ppm. The increase in the concentration of NO _x resulted in a linear increase in sensitivity for 10, 25, 50, 100, and 200 ppm of NO _x compared with the published study on WO ₃ sensors	[117]

Table 3 (continued)

Material	Gas sensed	Response	References
Porous rGO/WO ₃	NH ₃ gas	The three ratios (1%, 5% and 10% rGO/WO ₃) of nano-composite and pure WO ₃ showed good selectivity towards NH ₃ at 10–100 ppm, at room temperature in the range of about 32–35 °C. The pure WO ₃ sensor shows a response time of 165 s and recovery time of 132 s. However, the response and recovery times of the prepared sensor are 18 s and 24 s, respectively	[118]
GO/hexagonal(h)–WO ₃ nanorods	NH ₃ gas	The hexagonal structure of 1%GO. WO ₃ showed a better gas sensing performance for ammonia gas at 100 ppm ($R_a-R_g/R_g=17.6$) exceeding that of pure WO ₃ nanorods (1.27). The dominance of the lattice plane (221) in 1%GO. WO ₃ formed between GO and WO ₃ improved the electron transport in the gas-sensing process	[119]
rGO/WO ₃ nanowire	NH ₃ gas	rGO/WO ₃ -based sensor can detect highly toxic NH ₃ gas at low concentrations ranging from 20 to 500 ppm with detection limit of about 138 ppb	[120]
WO ₃ –Graphen	Detection of H ₂ and liquefied petroleum gas (LPG)	Composite film sensor with high sensitivity ($SF=4.5\times 10^{11}$) with an activation energy of 2.4 eV for H ₂ and low sensitivity for LPG ($\sim 2.0\times 10^{11}$) with an activation energy of 2.5 eV. Response time and recovery time were found to be ~ 1 s	[122]
WO ₃ /GN	Alcohol	At the operating temperature of 317 °C, the response of WO ₃ /GN nanocomposites shows for alcohol increases from 5.8 at 100 ppm to 40.9 at 2000 ppm with the highest rate of increase	[123]
rGO–h–WO ₃	Acetone	Materials behave as a p-type semiconductor after been exposed to reducing gasses. The response to 100 ppm of acetone of the rGO–h–WO ₃ sensor was observed to be 20% larger than the h–WO ₃ sensor. rGO/WO ₃ sensor exhibit the sensor response 1.55 at 200 ppm acetone concentration. The response time of the sensor was found to be 14 s. The specific surface area for gas adsorption was found 23 (m ² ·g ^{−1})	[124]
WO ₃ Nanolamella/rGO nanosheets	Acetylene	The higher sensor response (15.0 toward 50 ppm), faster response-recovery time (52 and 27 s), lower detection limitation (1.3 ppm)	[125]
WO ₃ –rGO	PAH (Anthracene)	The results showed that WO ₃ –rGO exhibited better sensitivity and accuracy compared to rGO with detection range between 10 pmol·L ^{−1} and 0.1 μmol·L ^{−1} of anthracene	[126]

photosensitizer, and incorporation of dopant that is vital for reinforcing the photoresponse of the visible light [129, 130]. WO₃ has a smaller bandgap, small light strength conversion efficiency, and the electron's discount ability due to the small bandgap, which is favorable for light absorption [36, 131]. Numerous reports in the literature have displayed better photocatalytic conduct of composites made up of tungsten trioxide and reduced graphene oxide compared to pure tungsten trioxide. The possible mechanism of dye degradation by WO₃/rGO composites is portrayed in Fig. 11. When WO₃/rGO is irradiated by visible light, it absorbs the incident photons, generating electrons and holes [43, 132]. The excited electron transfers from the conduction band (CB) of WO₃ to rGO sheets, which own a lower Fermi level ($E_F = -4.26$ eV)

and possess a better electrical conductivity (Eqs. (2) and (3)) [43, 102]. After that, the photogenerated electrons will be swept by dissolved oxygen (O₂) in water, resulting in effective electron–hole separation (Eq. (4)) producing $\cdot\text{O}_2^-$ [43, 132]. These $\cdot\text{O}_2^-$ will then turn to HOO $\cdot$, which will combine with the trapped electrons to generate H₂O₂, and finally produces OH $\cdot$ radicals (Eqs. (5)–(7)) [43]. Simultaneously, the holes left on VB of WO₃ can react with water and/or surface hydroxyl (–OH) to form hydroxyl radicals ($\cdot\text{OH}$) (Eqs. (8) and (9)). The radicals generated after a series of reactions attack the dye in the close vicinity, resulting in the degradation of dye and mineralization (Eq. (10)) [43, 132].

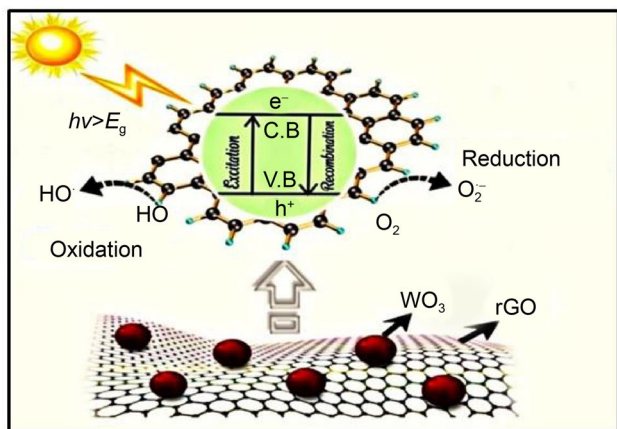
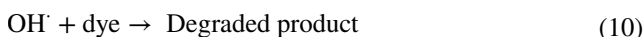
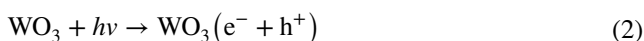


Fig. 11 Photocatalytic mechanism of WO_3/rGO



In the present review, the overview discussion on the nanocomposites WO_3/rGO -based photocatalyst, and its performance of WO_3 is presented and discussed in depth.

3.4.1 Degradation of dye and reduction of carbon dioxide

Dye degradation is a chemical reaction that breaks down large dye molecules into smaller molecules. Water, carbon dioxide, and mineral by-products are the end products which give the original dye its hue. Apart from their physical discomfort and toxicity, the ever-increasing mass production rate of dyes due to increased industrialization requires effective treatment [133]. As a result, photocatalysis is developed

to deal with complex effluents to reduce their potentially harmful environmental effects.

A simple chemical synthesis process and physical properties of WO_3 -rGO nanocomposites were reported in 2012 [134]. It was discovered that the presence of rGO in WO_3 -rGO composites facilitated electron transfer, light absorption, and electrical conductivity due to the significantly improved performance of WO_3 nanorods for the degradation of methyl orange (MO) under visible light irradiation as compared to pure WO_3 nanorods [134]. The photocatalytic property of the WO_3 -GO composite was also investigated by observing the degradation of methylene blue (MB) dye under ultraviolet (UV) light. GO doped WO_3 was synthesized using a template free deposition-hydrothermal route, showing high dispersity of GO onto WO_3 surfaces. Different carbon species were mediators that hindered the recombination rate of photogenerated electron-hole pairs and facilitated the electron transition [119].

A photoreduction method was successfully used to synthesize WO_3 nanorods @ graphene ($\text{WO}_3@\text{GE}$) nanocomposites [135]. The degradation of MO was used to test the photocatalytic activity of the $\text{WO}_3@\text{GE}$. Experimental results showed that the $\text{WO}_3@\text{GE}$ nanocomposites had higher photocatalytic activity. It was demonstrated that GE acted as an acceptor for the electrons generated in WO_3 and thus reduced the likelihood of photogenerated electron-hole pairs recombination [135]. The use of WO_3 nanoparticles in combination with rGO sheets synthesized via a simple one-pot hydrothermal method to degrade MB was published in 2015 [136]. Experimental results revealed that the superior contact between WO_3 and rGO sheets in the nanocomposites facilitates the electron transport and suppresses the electron-hole pair recombination in the nanocomposites [136]. For nucleation and growth of WO_3 particles, three-dimensional graphene foam (3D GF) was used as a highly porous conductive substrate. It was found that 3D graphene improved the photocatalytic properties of WO_3 due to the more straightforward and faster transfer of the photogenerated electron through the interconnected GR structure, the higher accessible surface area for WO_3 particle growth, and the easier adsorption of dye molecules [128]. A novel thermal reduction method was used to synthesize WO_3 -rGO nanocomposites that effectively catalyze the degradation of 1-naphthol under irradiation of a xenon lamp. It implies that the presence of rGO in the nanocomposites aided electron transfer. The kinetic studies showed the conformation of the experimental data to the first-order-kinetic model [132]. In 2016, graphene-incorporated WO_3 nanocomposites that initiate MB degradation under visible-light irradiation were reported [42]. It was discovered that incorporating graphene reduced the bandgap energy of pristine WO_3 while also acting as a charge carrier, effectively delaying electron/hole recombination and improving the photodegradation

performance of the $\text{WO}_3/\text{graphene}$ [42]. Visible light-driven photocatalysts WO_3/rGO were prepared via a facile one-step hydrothermal method to degrade sulfamethoxazole (SMX), a commonly used sulphonamide class antibiotic chosen as a target pollutant [137]. Compared with pure WO_3 , rGO-WO_3 composites showed significantly improved photocatalytic activity under visible light. It is proposed that increasing the catalyst loading would provide more available active sites for photocatalytic reactions and a large effective surface area for the adsorption process [137].

A simple hydrothermal method was used to prepare 3D nanostructured composite adsorbents of rGO and WO_3 (WO_3/rGO). Experimental findings revealed that WO_3/rGO exhibits excellent adsorption potential for residual Sr^{2+} from an aqueous solution over a pH range of 0 to 10 [138]. The data matched the Langmuir isothermal well ($R > 0.99$), and the maximum adsorption capacity of $149.56 \text{ mg} \cdot \text{g}^{-1}$ was achieved, which was higher than that of GO , WO_3 , and another related adsorbent. It is exposed that the rapid adsorption and high adsorption rate of WO_3/rGO are primarily due to the abundance of adsorption sites provided by the dispersed WO_3 nanoparticles on the rGO surface [138]. A series of composites containing hexagonal tungsten trioxide (h-WO_3) and rGO sheets are synthesized via a modified one-step hydrothermal route without assisted additive. Experimental results confirmed that h-WO_3 and reduced graphene have superior photocatalytic efficiency due to improved electrical conductivity of composites, which is helpful for the formation of e^- and the separation of an electron-hole pair [139]. A new and straightforward wet chemistry was followed by a thermal decomposition method to fabricate WO_3 plate-like and WO_3/rGO nanostructured catalysts [43]. It was discovered that incorporating rGO in the WO_3/rGO composite condensed the bandgap of WO_3 from 2.54 to 2.45 eV, inhibited the recombination rate of photogenerated electron-hole pairs, improved electron transport properties, and enhanced the degradation efficiency of MB compared to bare WO_3 [43]. One-pot synthesis of mesoporous WO_3 and $\text{WO}_3\text{-GO}$ nanocomposites employing the sol-gel method was reported in 2018 [140]. Platinum (Pt) nanoparticles were deposited onto WO_3 and $\text{WO}_3\text{-GO}$ to act as active sites for O_2 reduction, inhibiting electron-hole pair recombination in the $\text{Pt/WO}_3\text{-GO}$ nanocomposites. It was revealed that mesoporous $\text{WO}_3\text{-GO}$ nanocomposites with superior contact between WO_3 and GO sheets have outstanding photocatalytic activities, accelerating the rate of MB photodegradation [140].

In 2018, a viable strategy to fabricate WO_3/rGO composites via an in-situ solvothermal method was used for Photocatalytic purification of organic pollutants. It is exposed that the enhanced photocatalytic activity of the WO_3/rGO composite is primarily due to the formation of well-defined $\text{WO}_3\text{-rGO}$ interfaces, which effectively restrain the

recombination of photogenerated charges [141]. The visible light active photocatalysts WO_3 and WO_3/rGO were synthesized under the in-situ hydrothermal method to investigate the effect of catalyst quantity on photocatalytic contaminants as well as the effect of pollutant concentration on photocatalytic activity. Experimental findings highlighted that WO_3/rGO is the best active catalyst, and the simultaneous degradation of the pollutants is beneficial over the removal of individual contaminants [142]. GO -doping in WO_3 has been studied, proving that the small amount of GO (1%) does not effectively improve the electrostatic attraction between dye and catalyst [89]. The comparative investigations of WO_3/rGO thin film (wetting behavior and photocatalytic efficiency in Methylene blue removal) deposited by spraying stable dispersions of sol-gel powders were reported. Two distinct routes were used to synthesize the nanocomposites. It was discovered that the films obtained via Route 1 were more uniform, crystalline, and superhydrophilic while maintaining relative stability in photocatalysis conditions and comparable efficiency in methylene blue removal. Furthermore, the photocatalytic efficiency of thin films is reduced due to increased pollutant molecule adsorption on the photocatalytic layer [143]. Finding materials that aid in controlling water pollution caused by organic and bacterial pollutants is challenging. 2D sheets of WO_3 and composite of WO_3 and rGO have been synthesized well-controlled using a hydrothermal method. The greater degradation of Rhodamine B (RhB) dye was explained in terms of the molecular electrostatic potential. It was exposed that RhB has a more positive potential compared to MB dye where O_2^- and $\text{OH}^\cdot$ radicals interact more strongly, resulting in greater degradation of the RhB dye [46]. It has been reported that increasing the amount of WO_3 in GO increased its photocatalytic activity, and rapid photo-induced charge separation and blocking electron-hole pair recombination results in a higher separation efficiency of electron-hole pairs [144]. In addition, the increased visible light photocatalytic activity is due to the formation of heterojunction in which WO_3 is deposited on the smooth surface of GO , low bandgap, and generation of W-O-C linkages [144].

WO_3 nanorods are synthesized on the surface of graphene (GR) sheets using a one-step in-situ hydrothermal method employing sodium tungstate ($\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$) and GO as precursors. The photocatalytic activity of WO_3/GR nanocomposites under visible light was evaluated using methylene blue photodegradation, and the degradation rate of WO_3/GR nanocomposites was found to be twice that of pure WO_3 . This was recognized as the synergistic effect of graphene and WO_3 nanorods, which significantly improves photocatalytic performance, reduces recombination of photogenerated electron-hole pairs and increases visible light absorption efficiency [145]. In 2018, the outstanding photocatalytic efficiency of few-layer graphene- WO_3 was synthesized via

the arc discharge method. Experimental results revealed that the photocatalytic activity of WO_3 nanoparticles is increased by immobilizing them on graphene oxide sheets as graphene can easily trap the electrons excited in the valence band of WO_3 compared to pure WO_3 [146]. Optimization is essential in material synthesis as it affects the material's performance. It is a systematic way of an experimental design that can be accomplished by manipulating various physical and chemical parameters physical and chemical parameter of the experiment [147]. As determined by the Brunauer-Emmett-Teller (BET) analysis of pure WO_3 , the specific surface area of WO_3 ($17.61 \text{ m}^2 \cdot \text{g}^{-1}$) is lower than that of WO_3/rGO [148, 149]. However, as the weight ratio of graphene increases, the specific surface area increases, reaching a maximum of $143.87 \text{ m}^2 \cdot \text{g}^{-1}$ [149]. It can be concluded that the introduction of graphene is responsible for the high threshold performance of WO_3/rGO . On the one hand, excessive rGO after optimization promotes particle aggregation, blocks light absorption, and reduces light intensity arriving at the surface of the WO_3/rGO composite [150]. On the other hand, it also serves as a recombination center for electron-hole pairs [151], as proved by the Photoluminescence (PL), photocurrent, and Electrochemical Impedance Spectroscopy (EIS) studies [149].

In 2019, graphene- WO_3 nanocomposites with a different weight ratio of graphene were prepared by a hydrothermal method for synergistic degradation of enrofloxacin (EFA). It was revealed that graphene- WO_3 nanocomposites significantly improved the removal efficiency and first-order kinetic constant of EFA in the pulsed discharge plasma (PDP). Furthermore, the addition of WO_3/rGO could lead to the decline of ozone (O_3) and an improvement in OH and H_2O_2 [149]. To study the photocatalytic degradation of organic dyes such as Methylthionine chloride (MC) (cationic) and indigo carmine (IC, anionic), a WO_3/rGO nanocomposites were prepared using a simple ultrasonication mixing method. The photocatalytic degradation efficiency of WO_3/rGO nanocomposites and antibacterial activity is due to an improved electron-hole pair separation rate. The anticancer activity of WO_3/rGO nanocomposites was also tested on a human lung cancer cell line (A-549). It can be concluded that WO_3/rGO nanocomposites could be used as a foundation for developing photocatalytic applications [152]. In 2020, a two-step microwave thermal strategy was developed to synthesize a new hybrid catalyst comprising defective WO_{3-x} (where $0 < x \leq 1$, which is oxygen deficient, contains tungsten in a number of differing formal oxidation states) nanowires coupled with rGO to degrade persistent organic pollutants. It was discovered that interfacial WO_3 nanowires on rGO nanocomposites induced the localized super-hot spots on carbon materials which played a vital role in the structure formation and interfaced modification both in the liquid and solid phases. Furthermore, with the help

of an electron, mixed-valence $\text{W}^{4+}/\text{W}^{6+}$ promoted $\text{Fe}^{3+}/\text{Fe}^{+2}$ cycling, linking photocatalysis and Fenton reactions [153].

Recently in 2021, a series of WO_3/rGO nanocomposites [154] were prepared by changing the material ratio, the reaction temperature, and the reaction time and then added into a dielectric barrier discharge plasma (DBDP) system to investigate the degradation of bisphenol A (BPA) and the corresponding catalytic mechanism of the WO_3/rGO in the DBDP system. It was found that there was an optimum dosage of the WO_3/rGO ($40 \text{ mg} \cdot \text{L}^{-1}$) as well as the preparation conditions (5:1000 mass ratio of the GO and the WO_3 , 18 h reaction time, and 120°C reaction temperature) for achieving the highest catalytic effect [154]. It was also found that the WO_3/rGO shows much better visible light absorption and less photogenerated charge recombination than pure WO_3 due to the generation of crucial electron and hole pairs needed for the redox reactions which improves the rate of the photocatalytic activity [8]. The new function of graphene (GR) in boosting the two-electron reduction of O_2 to H_2O_2 was first discovered in the GR WO_3 nanorods prepared via a facile, solid, electrostatic self-assembly approach [155]. It has been found that graphene plays a vital role in improving the photoactivity of GR WO_3 NR nanocomposites in visible light; it increases the absorption of visible light and enhances the transfer rate of photogenerated charge carriers [155].

3.4.2 Reduction of carbon dioxide

It is high time now that we should understand the importance of reducing carbon footprint and harmful emissions and start taking steps to save our planet from global warming. Simulating the overall natural photosynthetic cycle of CO_2 chemical conversion into valuable fuels has been gaining popularity. Artificial photosynthesis allows for direct CO_2 conversion, lowering CO_2 concentrations in the atmosphere and providing green carbon fixation and energy storage [156].

In 2013, the fabrication of graphene WO_3 composites prepared via facile in-situ hydrothermal methods for photocatalytic CO_2 reduction was reported for the first time [156]. Graphene in graphene- WO_3 composites increased the conduction band of WO_3 for photocatalytic CO_2 reduction into hydrocarbon fuels when exposed to visible light. Experimental results point a new direction for developing effective photocatalysts and fine-tuning their photoreactivity for photocatalytic CO_2 conversion WO_3 to hydrocarbon fuels [156]. Table 4 summarizes the literature-reported applications of $\text{WO}_3/\text{graphene}$ -based nanocomposites as photocatalysts for dye degradation and reduction of CO_2 .

Table 4 Applications of WO₃/graphene-based nanocomposites as a photocatalyst reported in the literature

Material	Photodecomposition	Efficiency	References
WO ₃ -rGO	Degradation of MB	Without a catalyst: 6% With WO ₃ NPs: 47% ($k = 0.004 \text{ min}^{-1}$) With WO ₃ -rGO: 78% ($k = 0.008 \text{ min}^{-1}$)	[8]
WO ₃ /Graphene	Degradation of MB	WO ₃ /G-4: the highest degradation efficiency TOC removal by WO ₃ /G: 58%	[42]
rGO-WO ₃	Degradation of MB	Degradation ratio of MB observed in the presence of rGO-WO ₃ catalysts within 60 min was higher than that of bare WO ₃	[43]
rGO-WO ₃	Degradation of MB and RhB	MB No catalysts: 5% after 270 min WO ₃ and rGO-WO ₃ : 32% after 270 min RhB WO ₃ : 26% rGO-WO ₃ : 85%	[46]
GO-WO ₃	Degradation of MB	1% WO ₃ -GO: 100% WO ₃ : 34%	[119]
3D GF-WO ₃	Degradation of Rh B	GF-WO ₃ efficient than pure GF and WO ₃	[128]
WO ₃ /rGO	Photo degradation of 1-Naphthol	The degradation follows Langmuir-Hinshelwood kinetic law WO ₃ : 40% WO ₃ /rGO: 84%	[132]
WO ₃ -rGO	Degradation of MO	Degradation of MO under UV-light: Time dependent photo-reduction percentage: WO ₃ -rGO1: 37% 20 min WO ₃ -rGO2: 63% 40 min	[134]
WO ₃ @GE	Degradation of MO	WO ₃ @GE: 92.7%	[135]
WO ₃ -rGO	MB degradation	WO ₃ -rGO nanocomposites exhibit significant progress in photodegradation, compared to bare WO ₃ . It can be seen that the photocatalytic performance of the nanocomposite increased with increase in content of GO, and then decreased with the persistent increase of RGO content	[136]
WO ₃ -rGO	Degradation of SMX under visible light	Under visible light: SMX degradation (>98%) TOC removal efficiency: With WO ₃ : 35.7 With rGO: Increases to 57.4, 62.2, 67.9 Higher adsorption capacity with rGO/WO ₃ compared to neat WO ₃	[137]
RGO/WO ₃	Adsorption of Sr ²⁺	Adsorption capacity: Adsorption capacity is found to increase with increasing content of rGO	[138]
h-WO ₃ /rGO	Degradation of MB	Photocatalytic efficiency: m-WO ₃ : 63% WO ₃ -GO: 82%	[139]
Mesoporous WO ₃ and WO ₃ -GO	Degradation of MB	Mineralization of MB after 6 h, photo-efficiency: WO ₃ -GO: 80% Pt/WO ₃ -GO: 92%	[140]

Table 4 (continued)

Material	Photodecomposition	Efficiency	References
WO ₃ /rGO	Degradation of methylthionine chloride	TOC removal efficiency: WG0: 33.25% WG1: 38.62% WG2: 54.37% WG3: 58.75% WG4: 44.75% WG5: 32.87% The photocatalytic oxidation of p-cresol: Rate constants: WO ₃ : 0.0018 min ⁻¹ WO ₃ /rGO: 0.0023 min ⁻¹ Percentage of mineralization of p-cresol: WO ₃ : 68% WO ₃ /rGO: 79% The photocatalytic reduction of Cr(VI): Rate constants: WO ₃ : 0.0072, WO ₃ /rGO: 0.0084	Photocatalytic efficiency: WG0: 22.3% WG1: 29.5% WG2: 85.1% WG3: 94.1% WG4: 85.6% WG5: 57% [141]
WO ₃ /rGO	Degradation of p-cresol and Cr(VI)		[142]
WO ₃ -rGO	Methylene blue (MB)	Simultaneous removal of the pollutants is advantageous over the removal of the individual pollutant Route 1 proved more uniform, crystalline and superhydrophilic, while maintaining comparable stability in the photocatalytic conditions, and comparable efficiency in methylene blue removal: 25% after 8 h of irradiation with 10% UV + 90% VIS light, having the global irradiance of 34 W·m ⁻²	[143]
WO ₃ /GO	Degradation of MO	Pure GO: 50% 1.0% WO ₃ /GO: 70% 5.0% WO ₃ /GO: 75% 10.0% WO ₃ /GO: 95% 15.0% WO ₃ /GO: 77%	[144]
WO ₃ /GR	Degradation of MB	WO ₃ -G-1: 83% WO ₃ : 41%	[145]
Graphene-WO ₃	Degradation of MB	Few-layer graphene/WO ₃ with 15 wt%: highest efficiency	[146]
Graphene-WO ₃	Degradation of EFA	Graphene-WO ₃ nanocomposites significant improved the removal efficiency (99.1% with 60 min treatment in PDP system)	[149]
WO ₃ -GO	Degradation of MB and IC	Photocurrent density WO ₃ -rGO: 1.1 mA·cm ⁻² WO ₃ : 0.4 mA·cm ⁻² MB (%) WO ₃ : 78.96 WO ₃ -GO: 97.03	[152]
WO ₃ nanowire/rGO	Degradation of BPA	Optimized catalyst exhibited a remarkable performance in degradation of BPA with ~83% removal yield via photo-Fenton route	[153]
rGO-WO ₃	degradation of BPA	Highest degradation rate constant of the BPA was 0.03129 min ⁻¹ which was 1.78 times of the value achieved in the sole DBDP system and 1.68 times of that achieved in the DBDP/WO ₃ system	[154]
GR-WO ₃	Degradation of Rh B	Removal of RhB: 1% BGR-WO ₃ NR: ca. 96% WO ₃ NR: 14%	[155]
Graphene-WO ₃	Reduction of CO ₂	Yield of CH ₄ increased	[156]

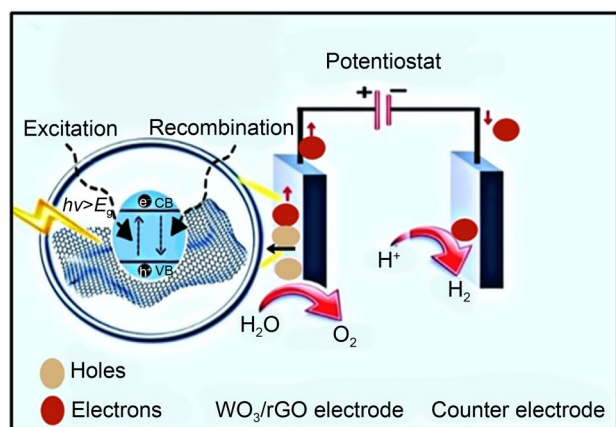


Fig. 12 Photoelectrocatalytic mechanism of WO_3/rGO

3.5 Photoelectrochemical application

Photoelectrochemical (PEC) cells are a particular type of solar cell that uses daylight and water to provide electricity and offers a hydrogen gas due to water splitting (Fig. 12). Solar energy can be converted into usable forms of energy with the help of PEC cells [19]. The term 'Photoelectrocatalytic' is used to represent processes associated with photocatalytic fuel cells. The popular reaction is (Eq. 11).



The splitting of H–O–H bonds in the water-splitting response can be accomplished via a variety of strategies, each of which differs from the others in terms of the energy source, as strength is necessary to break the bonds [157].

The semiconductor is crucial in a photocatalytic water-splitting reaction. When a semiconductor is exposed to visible light, electron–hole pairs are generated by the absorption of light of energy greater than or equal to bandgap energy [158]. The photogenerated electrons combine with hydrogen ions from dissociated water to form $\text{H}^\cdot$ radicals. Similarly, under the influence of the electric field, the photogenerated hole combines with the hydroxyl group to create $\text{OH}^\cdot$ radicals. $\text{OH}^\cdot$ radicals aid in the breakdown of waste products (such as dye), whereas $\text{H}^\cdot$ radicals result in the formation of hydrogen gas (H_2) [158, 159].

The study of developing efficient PEC substances based on transition metallic oxides has pushed water splitting as a sustainable strength asset for achieving the goal of solar electricity [157]. Among those potential candidates being substantially investigated, WO_3 has been considered a promising target for PEC anodes because of its enough bandgap ($E_g = 2.6\text{--}2.8\text{ eV}$) that can absorb light in the visible spectrum up to 500 nm and capture approximately 12% of the

solar spectrum [160]. The construction of hybrid structures among transition metal oxides and graphene is a promising method. The 2D graphene sheet has a large specific surface area and high intrinsic electron mobility to facilitate mild absorption, charge switch, and electrical conductivity [2]. Several reports in the literature [161, 162] show advanced PEC overall performance by incorporating graphene derivatives into WO_3 to enhance photoelectrochemical performance to overcome the inherent drawback of WO_3 as a photoelectrochemical material.

For the first time in 2012, the photo-electrocatalytic activity of tungsten oxide (WO_3) particles was synthesized on the surface of graphene (GR) sheets using a simple sonochemical method was reported. It was found that the chemical bonding between WO_3 and GR reduced the recombination of photogenerated electron–hole pairs, leading to enhanced photoconversion efficiency [161]. A simple UV-assisted reduction method successfully fabricated a novel visible-light-driven photocatalyst made of high-ordered mesoporous tungsten oxide and reduced graphene oxide. It was found that superior contacts between mesoporous WO_3 and rGO led to enhanced photo-electrocatalytic performance for O_2 evolution under visible irradiation. The enhancement of photocatalytic performance could be attributed to the large specific surface area, continuous meso-channels, and remarkable charge transfer and separation ability of the composites which provides a leading active site for photocatalytic reaction, providing a prototype for constructing a novel photocatalytic system by hybridizing graphene with mesoporous semiconductor for solar energy conversion [162].

It was also proved that the morphology of the GE- WO_3 nanowires clusters enables a multichannel environment to promote efficient charge interactions, resulting in reduced recombination of photogenerated electron–hole pairs and increased photocatalytic operation [135]. The WO_3 and WO_3/rGO composites were successfully synthesized by the simple, scalable, and green approach for water splitting and photocurrent generation application. The enhanced photoelectrochemical property owing to the incorporation of rGO in WO_3 improved the optical and electrical properties of the catalyst [163]. The fabrication of hybrid structures made of tungsten trioxide and graphene oxide via hydrothermal growth of ultrathin WO_3 nanoplates directly on fluorine-doped tin oxide (FTO) substrates was described. The increased photoactivity is attributed to the enhanced charge transfer caused by the incorporation of rGO, implying a particular approach for designing metal oxide–rGO hybrid architectures, according to electrochemical impedance spectroscopy [164]. A novel 2D/2D WO_3/rGO hybrid by coupling graphene sheets with WO_3 nanoplates has been fabricated that shows the improvement in photocatalytic efficiency of O_2 evolution due to the uncovering of electronegative oxygen atoms that result in the enhancement

of photo-induced charge carrier separation efficiency [98]. Fabrication of WO₃/rGO heterojunction electrodes with enhanced PEC (at 1.0 V vs Ag/AgCl under AM1.5 illumination) water splitting was reported for the first time. It has been discovered that rGO not only helps to separate electrons from holes by efficiently collecting and transporting photogenerated electrons, but also serves as an active site for electron–hole recombination, which is unfavorable for the electron–hole pair separation [165]. The photoelectrochemical property of WO₃ and WO₃/rGO photoanode coated on the Indium tin oxide (ITO) substrate was also reported. It was found that the WO₃ bandgap decreased from 2.54 to 2.45 eV with the addition of rGO, which impeded the recombination rate of photogenerated e[−]/h⁺ pairs and improved the properties of electron transport suitable for water splitting and photocurrent generation applications [43].

The development of Nanostructured WO₃/rGO composite has been prepared hydrothermally, and their PEC activities under different circumstances, i.e., dark and light environments, were reported. Experimental results revealed that the photoresponse of electrodes appears in the form of a tremendous amount of photocurrent against applied voltage during each electrode's investigation. Optimal graphene concentration can cause perfect photocatalytic activity. Additionally, the optical bandgap of prepared composites is a little high and directly affects the efficiency of prepared electrodes. Further studies of these hybrid WO₃/rGO nanocomposites structures, with rational bandgap and doping design, may lead to various opportunities for optimizing transition metal oxide-based PEC conversion of solar energy [166]. Table 5 depicts the photo-electrocatalytic performance of WO₃/graphene-based nanocomposites.

3.6 Application in the medical field

Tungsten oxide nanostructure has been considered a promising material as an immobilized (to reduce or eliminate motion of (the body or a part)) matrix due to its enhanced catalytic activity, chemical stability, simple synthesis, and robust adherence to the substrate [167]. The graphene-based nanocomposites are considered a promising support matrix to boost the biosensing characteristics [168]. A few reports in the literature may be due to a lack of attention demonstrating stepped forward antibacterial activity and biosensing characteristics by incorporating Graphene into WO₃.

The antimicrobial activity of WO₃/rGO nanostructure against Gram-positive microbes (*B. Subtilis*) and Gram-negative (*P. aeruginosa*) was reported in 2017 for the first time [46]. A possible explanation for the antimicrobial activity of the Tungsten oxide nanostructures against the *B. Subtilis* and *P. aeruginosa* bacterial strains is an electrostatic interaction between the positive Tungsten oxide nanostructures and the negative charges of the microbes [169]. It could also be

because the ions released by semiconducting nanomaterials react with the proteins present in the bacterial cells, which leads to the death of the microbes [169, 170]. In 2018, the WO₃/rGO nanocomposites were used as an immobilized matrix for detecting the cardiac biomarker cardiac troponin I (cTnI) [167]. The experimental results revealed that heterogeneous electron transfer kinetics of WO₃/rGO nanocomposites led to improved immunosensor efficiency.

Volatile organic compounds (VOCs) sensing properties of the bumpy, high surface area WO₃ hemitubes functionalized by graphene-based compound (Fig. 13) for the diagnosis of diabetes and halitosis (A persistent, unpleasant odor in exhaled breath, typically not severe, generally known as bad breath) was reported first time [121]. The superior sensing properties were attributed to the electronic sensitization of graphene-based materials by modifying space-charged layers at the interfaces between n-type WO₃ hemitubes and p-type graphene-based materials, as supported by Kelvin probe force microscopy (KPFM). In 2019, the fabrication of graphene/tungsten trioxide (Gr/WO₃) composite modified Screen-printed carbon electrode for detection of nitrofurantoin (NTF) was reported [168] as overdose of NTF causes mutagenicity, hepatotoxicity [171, 172] and carcinogenic activity [173]. It is also revealed that the electrochemical behavior of NTF is a pH-dependent electrochemical reaction. Testing the immunosensor with cardiac patient samples indicates the process of biosensing clinical application to detect other biomarkers. The most recent diagnostic and halitosis sensing materials could be developed by monitoring the number of graphene-dependent additives and the thermal aging stage [121]. The various applications of WO₃/graphene-based nanocomposites in the medical field are shown in Table 6.

4 Present scenario

WO₃/graphene-based nanocomposites have demonstrated excellent performance as electrochromic materials, supercapacitors, Li-ion batteries, gas sensors, photocatalysts, as well in the medical field. Figure 14 depicts the contribution of different areas last ten years. The most significant contribution comes from photocatalysis (31.8%), as photocatalysis holds the promise of resolving issues related to the intermittent existence of sunlight, which is regarded as a sustainable and ultimate energy source for powering activities on Earth, followed by gas sensors (27.1%). Supercapacitors and lithium-ion batteries contribute 18.8%.

To expand the light absorption range, accelerate the interfacial charge transfer, reduce the electron/hole recombination, and improve the overall performance, ternary composites associated with the WO₃–graphene can show apparent advantages over the binary composites for the synergistic

Table 5 Photoelectrocatalytic performance of WO₃/graphene-based nanocomposites reported in the literature

Material	PEC activity	Result	References
rGO–WO ₃	Evolution of hydrogen	On incorporation of rGO, the bandgap of WO ₃ reduced from 2.54 to 2.45 eV. The increased photon-to-hydrogen conversion efficiency of 5.3 and 4.2 folds at the lower bias potential of 0.08 and 0.30 V respectively by 1.0 GW photoanode compared to WO ₃ photoanode	[43]
2D/2D rGO/WO ₃	O ₂ evolution	The 3 wt.% rGO/WO ₃ heterojunctions display significantly the increase of O ₂ production, which is 3.9 and 7.3 times higher than those of WO ₃ nanoplates with preferential (0 0 2) facets and commercial WO ₃	[98]
GE–WO ₃	Evolution of oxygen	Amount of O ₂ evolved after 8 h from water splitting: Bare WO ₃ : <i>ca.</i> 0.175 mmol·L ⁻¹ GE–WO ₃ (2.0 wt% G.E): <i>ca.</i> 0.345 mmol·L ⁻¹	[135]
WO ₃ @GR	Evolution of oxygen	Amount of O ₂ evolved after 9 h: WO ₃ @GR – <i>ca.</i> 338 μmol·L ⁻¹ Mixed WO ₃ /GR – <i>ca.</i> 214 μmol·L ⁻¹ WO ₃ – <i>ca.</i> 186 μmol·L ⁻¹	[161]
Graphene–WO ₃	Evolution of hydrogen	Rate of H ₂ evolution: 7% graphene–WO ₃ : 288 μmol·h ⁻¹ ·g ⁻¹ Photocatalytic hydrogen evolution efficiency rises by increasing graphene content	[162]
WO ₃ /rGO	Evolution of oxygen	Photocurrent density: 2 wt % rGO: 4.0 μA·cm ⁻² 6 wt% rGO: 13.0 μA·cm ⁻² Oxygen evolution form water after 5 h irradiation: m-WO ₃ : 85.0 μmol·g ⁻¹ 2 wt% rGO: 182.4 μmol·g ⁻¹ 6 wt% rGO: 232.4 μmol·g ⁻¹ 0.5% wt Pt m-WO ₃ /rGO: 437.3 μmol·g ⁻¹	[151]
rGO–WO ₃ composite	Evolution of hydrogen	rGO–WO ₃ composites showed the enhanced photoelectrochemical hydrogen conversion efficiency from 0.6% (for WO ₃) to 2.7% and a twofold increase in photocurrent generation under solar light irradiation. The photocurrent (at 1.23 V applied potential) of all the rGO–WO ₃ composite photoanodes are higher than that of pure WO ₃ photoanode	[163]
WO ₃ /reduced graphene oxide	Evolution of hydrogen	WO ₃ /rGO nanocomposite have achieved a photocurrent density of 2.0 mA·cm ⁻² at 1.23 V vs. reversible hydrogen electrode (RHE)	[164]
WO ₃ –RGO	Evolution of hydrogen and oxygen	Water splitting PEC system Time for evolution: 4 h H ₂ evolution WO ₃ –rGO: 60 μmol·cm ⁻² WO ₃ : 25 μmol·cm ⁻² O ₂ evolution WO ₃ –rGO: 27 μmol·cm ⁻² WO ₃ : 12 μmol·cm ⁻²	[165]
WO ₃ /rGO	Evolution of hydrogen	Photocatalytic hydrogen evolution efficiency rises by introducing rGO. The electrodes exhibit photocurrent ranging from 20 μA to 2.638 mA under a dark environment. The photoresponse of electrodes appears in the form of a huge amount of photocurrent against applied voltage	[166]

effect of multi-components [174]. In the ternary composites associated with the WO₃-graphene, graphene can serve as a conductive substrate to support metal nanoparticles or metal oxides. Furthermore, the formation of a heterojunction on

graphene efficiently accelerates the interfacial charge transfer and suppresses the recombination of electron–hole pairs. However, the selection of the components, microstructure design, and the optimization of ratios decide the overall

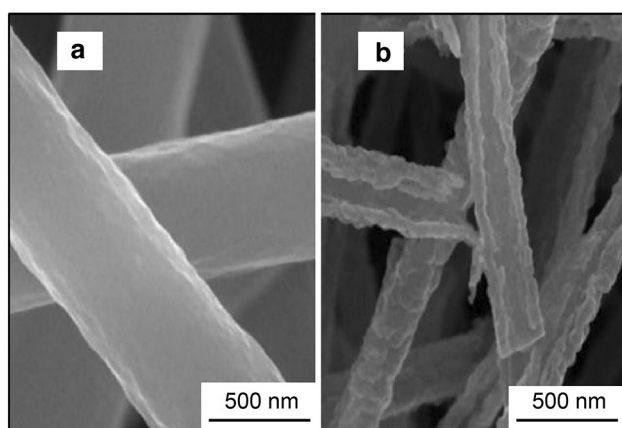


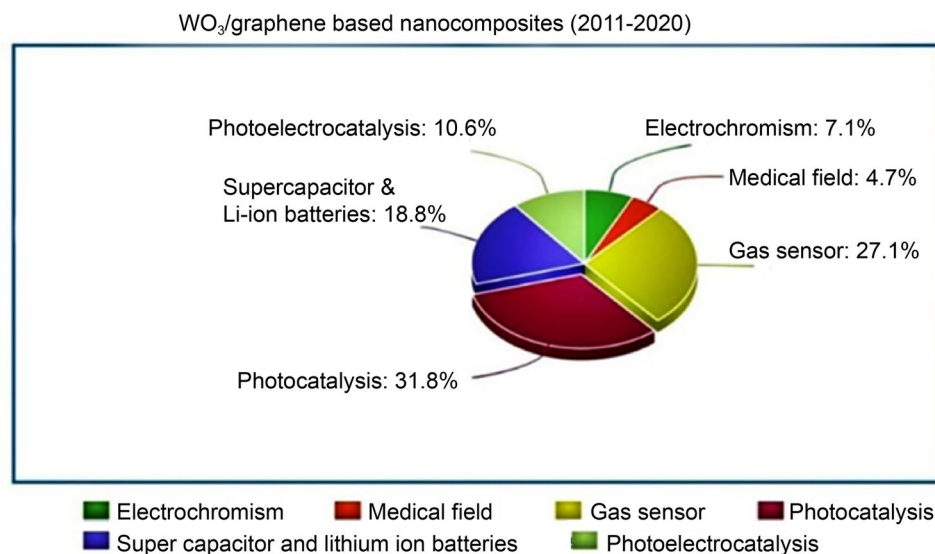
Fig. 13 **a** as-spun fibers **b** bumpy WO_3 hemitubes after calcination at $500\text{ }^\circ\text{C}$ for one hour from Ref. [121]. Reproduced with permission from Ref. [121]. Copyright 2014, American Chemical Society

performance of ternary composites [174]. For instance, to increase the interfacial contact between TiO_2 and WO_3 , a Z-scheme heterojunction-based ternary composite of $\text{TiO}_2/\text{rGO}/\text{WO}_3$ (TRW) was formed by depositing TiO_2 nanocrystals and WO_3 nanorods onto rGO substrate. It has been discovered that rGO successfully inhibits electron–hole pair recombination and speeds up electron transport and O_2 reduction processes. As a result, as compared to the TiO_2/WO_3 combination, the ternary composite had better bacterial inactivation ability [175]. A salt ultrasonic-assisted hydrothermal process was also used to make a ternary composite of spherical TiO_2 and WO_3 nanoparticles on rGO sheets. The ratio of $\text{TiO}_2/\text{WO}_3/\text{rGO}$ was optimized to achieve a larger specific surface area with the best photocatalytic activity for the destruction of RhB [176]. A one-step hydrothermal technique was also used to fabricate a ternary composite of graphene-modified WO_3/TiO_2 step-scheme (S-scheme) heterojunction. The development of S-scheme heterostructures and Schottky junctions in this ternary composite accelerates charge transfer and prevents electron and hole

Table 6 Various applications of WO_3 /graphene-based nanocomposites in the medical field reported in the literature

Material	Result	References
rGO– WO_3	The rGO– WO_3 composites act as better disinfectants compared to bare WO_3 . A higher antibacterial effect was observed for Gram-positive microbes (<i>B. subtilis</i>) compared to Gram-negative (<i>P. aeruginosa</i>) microbes	[46]
GO– WO_3	GO– WO_3 nanocomposite exhibited fast response ($< 15\text{ s}$) and recovery ($< 30\text{ s}$) toward acetone H_2S and showed the excellent potential for real-time exhaled breath diagnosis of diabetes and halitosis	[121]
WO_3/rGO	Immunosensor exhibits a good sensitivity of $58.24\text{ }\mu\text{A}\cdot\text{cm}^{-2}$ per decade in a wide detection range of $0.01\text{--}250\text{ ng}\cdot\text{mL}^{-1}$ with stability up to 30 d as well as good reproducibility and excellent selectivity toward the detection of cTnI biomarker	[167]
Graphene/ WO_3	Graphene/Tungsten trioxide (Gr/ WO_3) composite modified screen-printed carbon electrode shows an excellent electrocatalytic activity toward NTF with a lower detection limit ($0.002\text{ }\mu\text{mol}\cdot\text{L}^{-1}$), well linear response range ($0.01\text{--}234\text{ }\mu\text{mol}\cdot\text{L}^{-1}$) and acceptable sensitivity ($2.18\text{ }\mu\text{A}\cdot\text{L}\cdot\mu\text{mol}^{-1}\cdot\text{cm}^{-2}$)	[168]

Fig. 14 Statistical study based on WO_3 /graphene's nanocomposite for various applications last ten years



recombination, enhancing photocatalytic H₂ evolution performance [177].

5 Outlook and future scope

Although WO₃/graphene-based sensors, supercapacitors, electrochromic devices, and photocatalytic electrodes possess several advantages, they still have challenges and difficulties. As wearable/flexible devices based on WO₃/graphene are still in great demand. But, the fundamental understanding of the relationship between microstructure and overall performances must be explored in theory. The conductivity and PEC activity of WO₃ via rGO incorporation in WO₃/rGO heterojunction photoanode significantly increased. Nevertheless, further research is required to make a system for H₂ production at the commercial level. In the realm of electrochromic, key issues are: modeling of the diverse range of devices, novel device designs for specific applications, and performance optimizations and device stability.

Electrochromic devices based on WO₃/graphene are broadly explored as variable emissivity modulators. But still, they exhibit several drawbacks: they cannot bend or twist, have slow switching speed, and challenge material processing. The hybrids of WO₃ and graphene for near-infrared region (NIR) shielding applications have yet to be fully explored. WO₃/graphene-based nanostructures as electrode materials are expected to display enhanced electrochemical performance because of their large surface area and low charge transport resistance. However, low conductivity and poor rate performance limit its broad application of pseudocapacitors at the commercial level. For gas sensing applications, the comprehensive performances are still far from practical applications such as higher working temperature, low selectivity, and long-term stability. Besides, a deep understanding is urgently expected to explore the enhanced mechanism.

Moreover, leaching tests should be performed to evaluate the components' strength and lack of lixiviation when these materials are tested for water treatment. One of the main limitations in the medical field is the failure to demonstrate enhanced antibacterial activity and biosensing characteristics by incorporating graphene into WO₃. Therefore, a new research effort is required to understand the mechanisms and factors that influence the interaction between nanocomposites and microorganisms. The advantages of both electrochemical and optical approaches to PEC sensors could also be achieved by combining electrochemical detection with light irradiation for prostate detection. And photoelectrochromism can be realized by linking solar cells with an ECD for storage for an efficient change in the transmittance of visible and near-infrared light without any external

power source. For electrochromism, colouration efficiency is critical. It can be enhanced using WO₃/graphene and metal nanoparticles to make the ternary composite. The variable emissive properties electrochromic device can work as an ideal spacecraft antenna for the thermal management system. As satellites and spacecraft have an arduous task of thermal management because of the high-temperature changes involved in space. This can be achieved by covering satellite surfaces with VE coatings. It is necessary to address these problems and enhance the performance of ECD devices for wide-area applications.

6 Conclusion

The progress made in the last ten years using WO₃/graphene-based nanocomposites was summarized in this paper. Composites based on WO₃/graphene can be nanoscale and utilized for the advantage of human health, environmental protection and remediation, and power conversion to make a brand new global in the future. This review paper encompasses a detailed study of preparation methods of WO₃/rGO-based nanocomposites. The strategies and importance of sandwich structure to improve the performance of binary composite have also been discussed. Electrochromic windows employ the structural and functional features of WO₃ and graphene derivatives used at commercial levels. The WO₃/rGO composite is used as an antibacterial material, a sensory platform for RZ detection, and an electrochemical detection platform for the human cardiac biomarker Troponin I (cTnI) have all been addressed.

Moreover, the benefits of ternary composites containing WO₃-graphene over binary composites have also been highlighted. The WO₃/rGO composite green electrode complements capacitive properties with higher reversible charging/discharging potential and better capacitance values than those of the tungsten oxide electrode and provides a viable path for future research. Nanocomposites based on WO₃-graphene exhibit excellent properties, but recombination of the electron-hole pair in tungsten oxide limits their application as photocatalysts. In the future, Z-schemes with ternary heterojunctions may offer better charge separation due to ternary heterojunctions. Therefore, researchers are exploring new ways to overcome the limitations of tungsten oxide with graphene derivatives, particularly for applications in photocatalysis. Excellent absorption behaviors of WO₃/rGO composite toward natural compounds in wastewater due to their beautiful backgrounds consisting of nontoxicity, immoderate photocatalytic degradation capacity, and thermal and chemical stabilities are also discussed. This review article provides the research gap and can excite new ideas to defeat the associated challenges for further improvement of graphene-based tungsten oxide nanocomposites.

Acknowledgements This research was partially supported by The Institute of Science, Dr. Homi Bhabha State University, 15, Madam Cama Road, Mumbai-400032, MH, India.

Declarations

Conflict of interest The authors declare no conflict of interest.

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